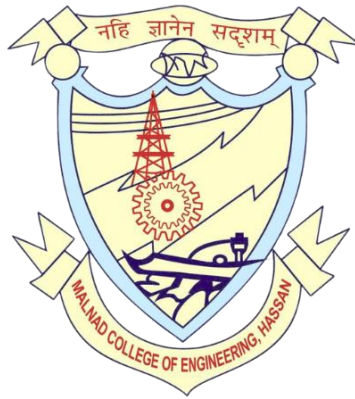


MALNAD COLLEGE OF ENGINEERING

(An Autonomous Institution Under VTU, Belgaum)

Department of Computer Science and Engineering



Module 1

Course Title: Computer Networks

Course Code : 24CS503

Module1

Introduction: Data Communications, Network Types, Network Models: TCP/IP Protocol Suite. Introduction to Physical Layer: Data and Signals, Periodic Analog Signals, Digital Signals, Transmission Impairment, Data Rate limits, Performance. Digital Transmission: Digital-to-Digital Conversion - Line Coding, Line Coding Schemes.

Introduction

- Data Communications basics
- Network – Criteria and Structures
- Network types
- Internet History
- Standards and Administration

Data Communications

Communication

- **Sharing** of information – local/ remote (tele - far)

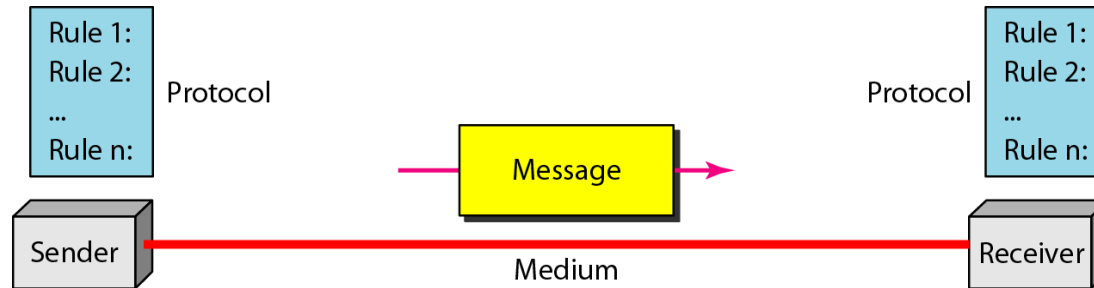
Data Communication

- Exchange of data between two devices via some transmission medium
- Data - information presented in whatever form is agreed upon by the parties

Fundamental Characteristics

1. **Delivery** – Correct destination, Only intended device/user
2. **Accuracy** – Data must not be altered during transmission
3. **Timeliness** – Data must be delivered without delay (Real-time)
4. **Jitter** – Variation in delays – should be minimum

Communication System - Components



1. **Sender** – Device that sends the message
 - Telephone, Computer, ...
2. **Receiver** – Device that receives the message
 - Telephone, Computer
3. **Message** – Information (data) that is communicated
 - Text, Numbers, Pictures, Audio, Video
4. **Transmission Medium** – Physical path
 - Copper Wires, Coax, Radio waves, Optical fiber
5. **Protocol** – Set of rules that govern data communication

Data Representation

1. **Text** – ASCII (8-bit), Unicode (32-bit)
 2. **Numbers** – Binary
 3. **Images** – A matrix of picture elements (pixels)
 - Each pixel is assigned a value
 - Gray Scale – 8 bits
 - Colour– 3 Bytes (RGB or CMY)
 4. **Audio** – Speech/ Music - Digitized
 5. **Video** – Sequence of images
- **Analog** signals are converted into **digital** form at the sending end - They may be converted back to analog form at the receiving end.

Data Flow

1. Simplex – Unidirectional

- Keyboard, Monitor, Printer, Radio and Television broadcasting, PA System

2. Half Duplex – Bidirectional but one at a time

- Walkie-Talkie, Citizen Band Radio

3. Full-Duplex (Duplex) - Bidirectional and simultaneous

- Telephone

1.A university launches a Smart Learning Platform that offers recorded lectures, live online classes, and virtual office hours. During recorded lectures, students only receive content. During live classes and office hours, students and instructors exchange audio, video, and chat messages simultaneously. As a network engineer, analyze the communication requirements of each service and identify the most appropriate communication mode.

Answer:

Recorded Lectures → **Simplex**

Live Online Classes → **Full-Duplex**

Virtual Office Hours → **Full-Duplex**

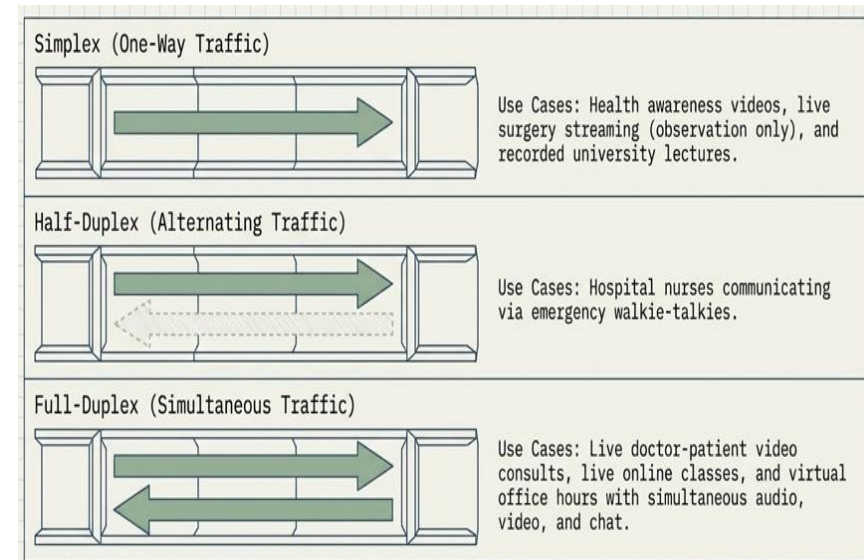
2.A multi-specialty hospital uses a Digital Healthcare System where patients watch health awareness videos on digital screens, doctors conduct live video consultations with patients, nurses communicate using walkie-talkies during emergencies, and surgeries are live-streamed to medical students for observation only.As a Network Engineer, analyze the communication requirements of each service and identify the most appropriate communication mode.

Answers:

Simplex: Health awareness videos, Live surgery streaming

Half-Duplex: Walkie-talkie communication

Full-Duplex: Live doctor-patient video consultations



Networks

A network is a set of devices (nodes) connected by communication links, capable of communication

Network Criteria

1. Reliability

- Robustness. No or very less failures

2. Security

- Protection to data from unauthorized accesses and data losses

3. Performance

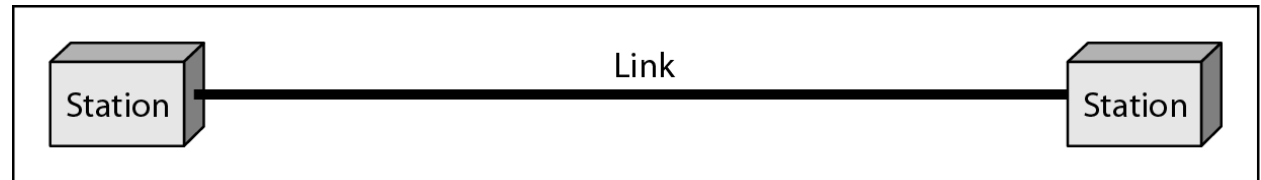
- Transmission time (Transit time), Response Time,
- **Measures:** Throughput and Delay

Networks - Physical Structures

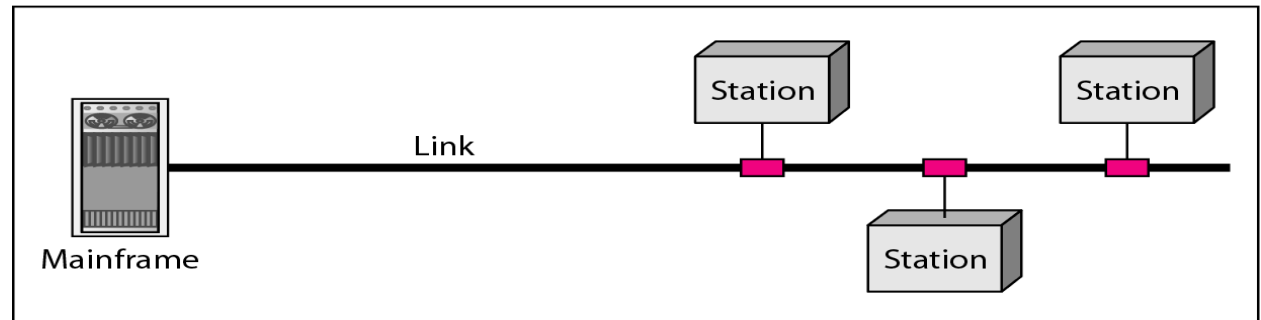
Types of Connection

- **Point-to-Point** : Dedicated link between devices
- **Multi-point (Multi-drop)**: Shared link

Point-to-Point



Multi-point



Networks - Physical Structures

Topology: Physical way in which the devices are interconnected

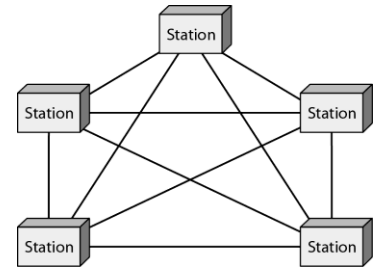
- **Regular:** Mesh, Star, Bus, Ring
- **Irregular :**Using Point-to-point links
- **Hybrid :**E.g. Main topology – Star, Each branch - Bus

Network Topologies - Mesh

- Each node has a dedicated **point-to-point** link to every other
- If there are n nodes $n(n-1)/2$ links are required (duplex)

- **Advantages**

- No traffic problems
- Robust – A link failure affects only two nodes
- Privacy and Security
- Fault detection and repair are easy



- **Disadvantages**

- More I/O Ports and Cables required. As n increases the total number of links required increases enormously
- Installation and maintenance are difficult
- Very expensive

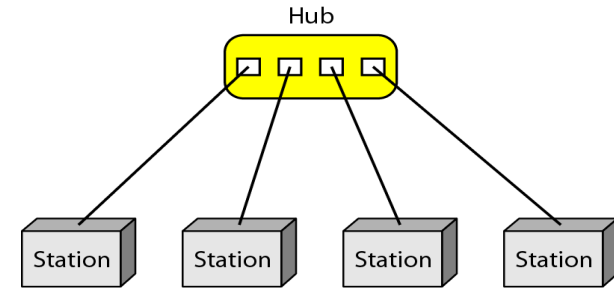
- **Example** – Interconnection of telephone regional offices

Networks Topologies - Star

- Each device has a **point-to-point** link to a central controller – Hub
- Hub relays the message to the desired destination
- If there are n nodes n links are required

Advantages

- Less number of links required
- Addition of a node requires only one additional link
- Robust – A link failure affects only one node
- Less expensive
- Installation and maintenance is easy



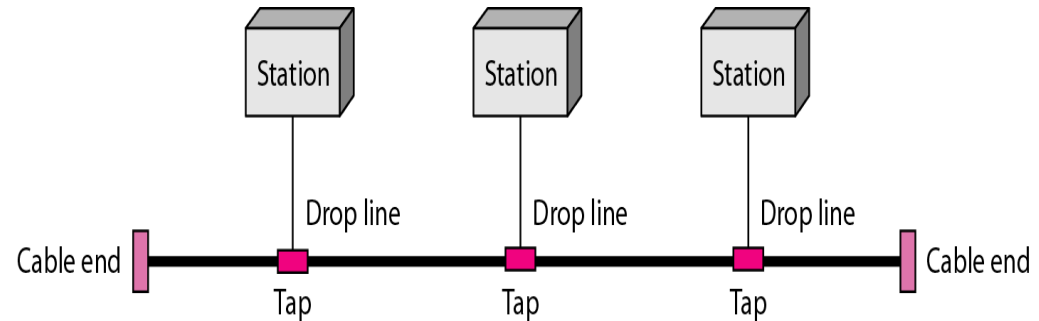
Disadvantages

- Hub becomes a critical component
- Cabling is more compared to some other topologies

Example – LANs

Networks Topologies - Bus

- **Multi-point** - Only one common bus
- Nodes are connected to the bus using taps and drop lines
- **Advantages**
 - Less cabling
 - Easy installation
 - Less expensive
- **Disadvantages**
 - Fault in the cable affects all nodes
 - Fault detection is difficult
 - Cable length and number of taps are limited
- **Example** – Ethernet LANs - Not widely used now.

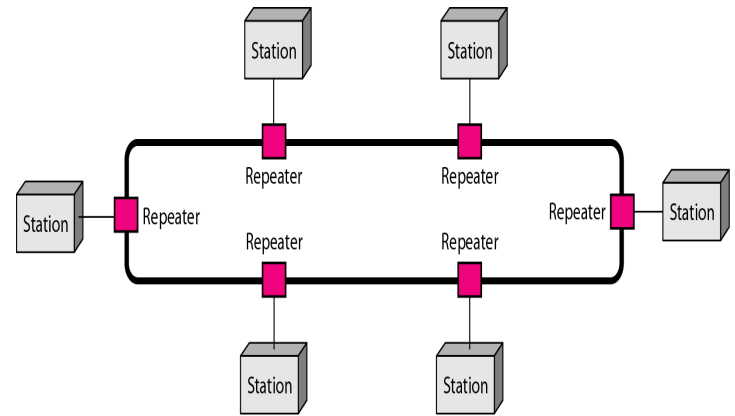


Networks Topologies - Ring

- Each device has a point-to-point connection with two devices on either side of it
- Each node is connected to a repeater on the ring
- Signal is passed in one direction

Advantages

- Less cabling
- Easy installation
- Less expensive
- Addition of a new node requires only two connections



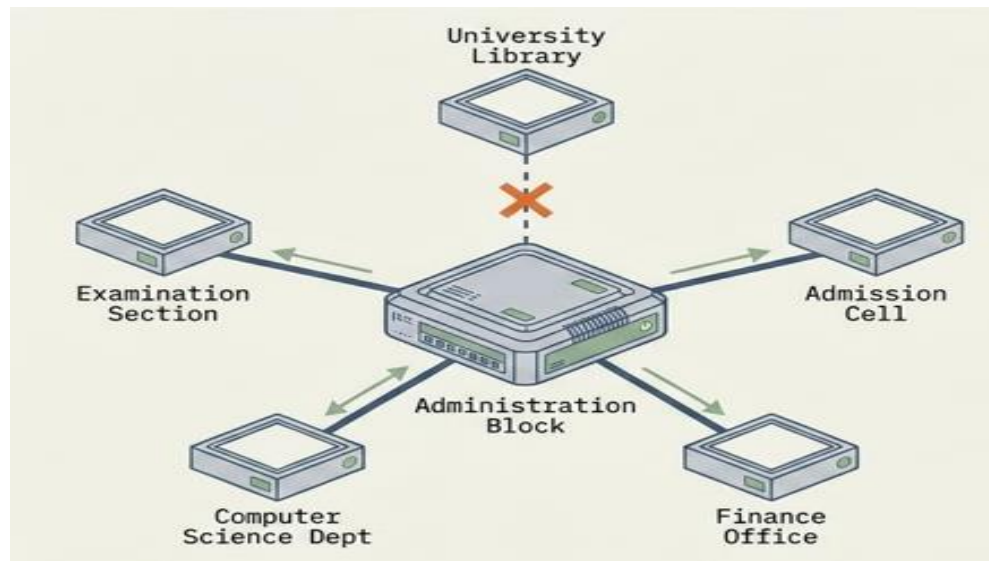
Disadvantages

- Length of the ring and number of nodes are limited
- Fault detection is easy
- A break in the ring affects the entire network

Example – IBM's Token Ring LANs - Not widely used now

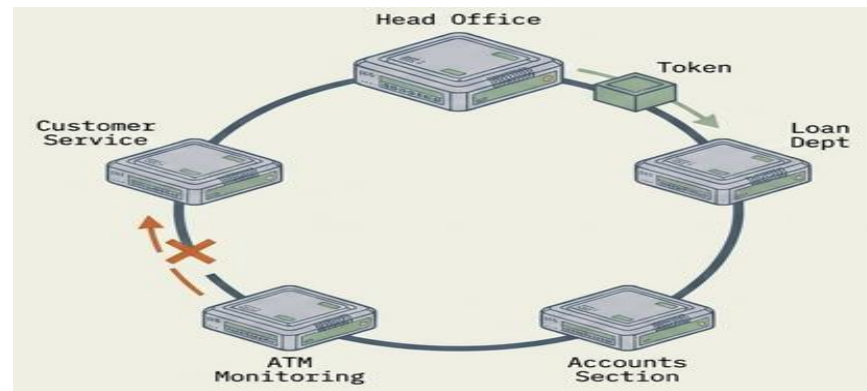
1. A university is establishing a campus-wide network to connect the Administration Block, Library, Examination Section, Computer Science Department, and Research Center. The network administrator wants a design that simplifies management, allows easy addition of new departments in the future, and ensures that a failure in one department does not affect communication among the others. As a network engineer, recommend a suitable topology and justify your choice based on reliability, scalability, and maintenance requirements.

Answer:



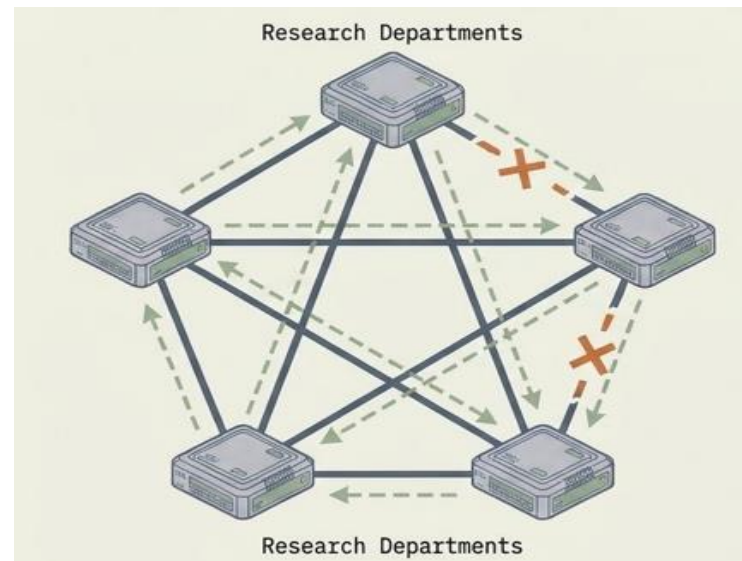
2. A national bank has interconnected its Head Office, Loan Department, Accounts Section, ATM Monitoring Centre, and Customer Service Department in a Local Area Network (LAN). During business hours, all departments continuously exchange financial data. To prevent data collisions and ensure that every department gets an equal opportunity to transmit data, the network uses a token-passing mechanism. However, the management is concerned about the effect of a communication link failure on the network. Identify the most suitable network topology for this scenario. Explain how data is transmitted using the token-passing mechanism. Also, analyze the advantages and limitations of this topology in terms of reliability, performance, and fault tolerance.

Answer:
Ring Topology



3. A university research campus contains five departments that frequently exchange confidential research data. The management requires direct communication between every pair of departments to maximize reliability, security, and fault tolerance. Even if one communication link fails, data transfer between other departments should continue without interruption. Which network topology would you recommend? Analyze its advantages and limitations in this scenario.

Answer:



Practice question:

1.The University Library, Examination Section, Admission Cell, and Finance Office provide critical services that must remain available throughout the academic year. The network designer must choose between Star and Bus topologies. A network failure should affect the minimum possible number of offices, and troubleshooting should be completed quickly. Analyze both topologies and recommend the most suitable one for this environment.

Types of Networks

1. Local Area Network – LAN

2. Wide Area Network – WAN

- **Metropolitan Area Network - MAN**
- **Room Area Network – RAN**
- **Personal Area Network – PAN**

Local Area Network - LAN

- Normally covers an area of 1-2 Kms radius
 - Office, Building, Campus
- Usually privately owned
- High-speed communication
 - Data rate - 10 Mbps, 100 Mbps, 1000 Mbps (Gigabit)
- Low error rates
- Topologies – Bus, Ring, Star
- Media – Only one type
 - Twisted pair, Thin coax, Wireless
- Low cost
- Sharing of resources – Hardware/ Software

Wide Area Network - WAN

- Covers very large areas – can be worldwide
- Usually owned by service providers (ISP)
- Low-speed communication (< 100 Mbps)
- Higher error rates
- Topologies – Irregular

Types

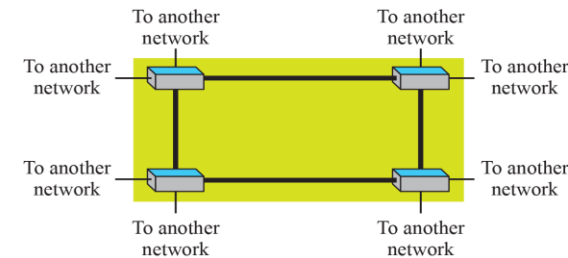
• Point-to-Point WAN

- Network that connects 2 devices
- E.g. Dial-up line to Internet



• Switched WAN

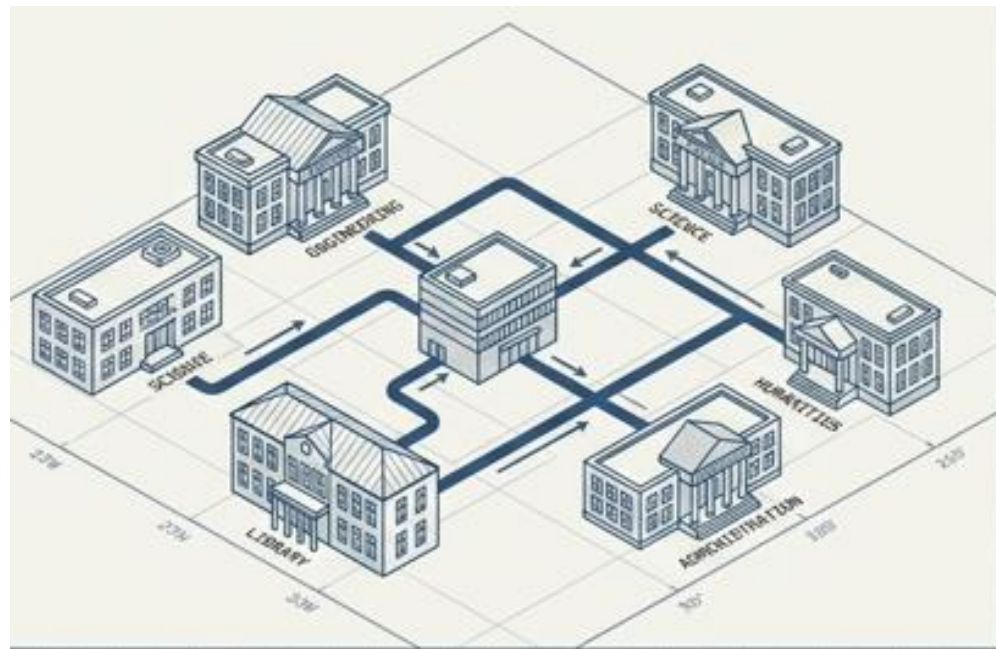
- Back-bone that makes the Internet
- Several PPP WANs connected by switches
- X.25, Frame Relay, ATM (Asynchronous Transfer Mode)



1. A university has multiple departments located within a 2 km radius. Which type of network would you recommend to connect all departments and why?

Answer:

LAN (Local Area Network)



2. A university student uses a laptop to attend online classes, a smartphone to receive notifications, a smartwatch to monitor health data, and wireless earbuds for audio communication. The student wants all devices to exchange data seamlessly within a short range without relying on a campus-wide network. Which type of network would you recommend and why?

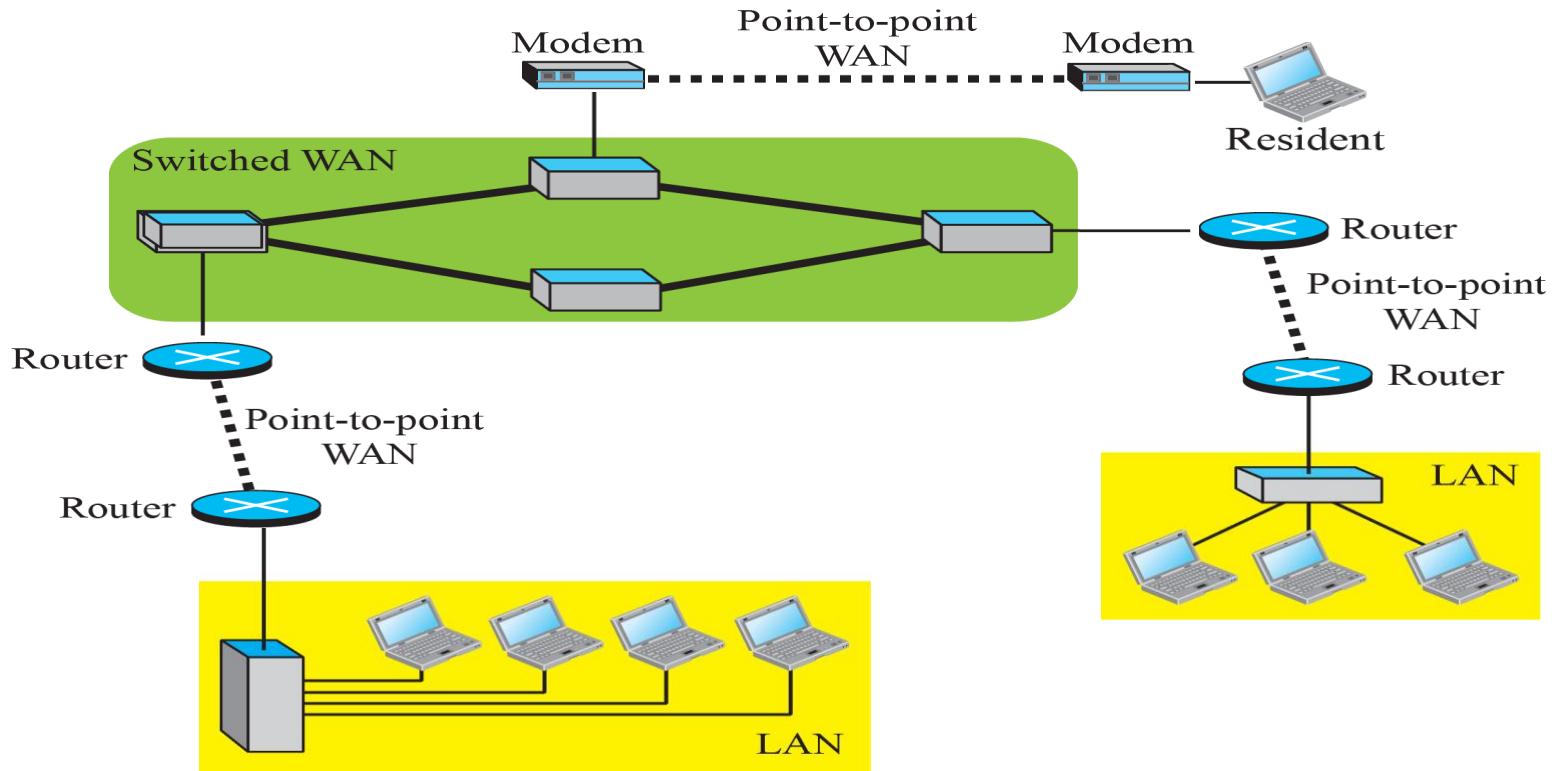
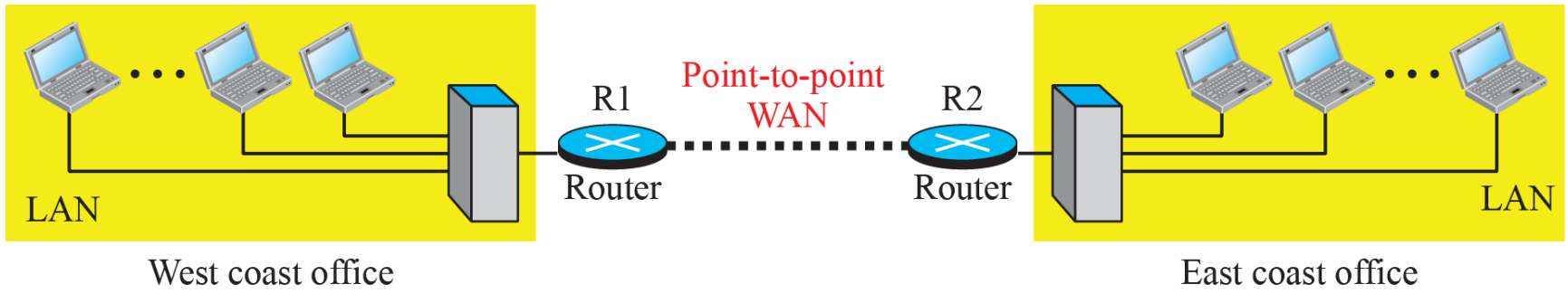
Answer: PAN because it is designed to connect personal devices within a short range, typically using Bluetooth or similar wireless technologies



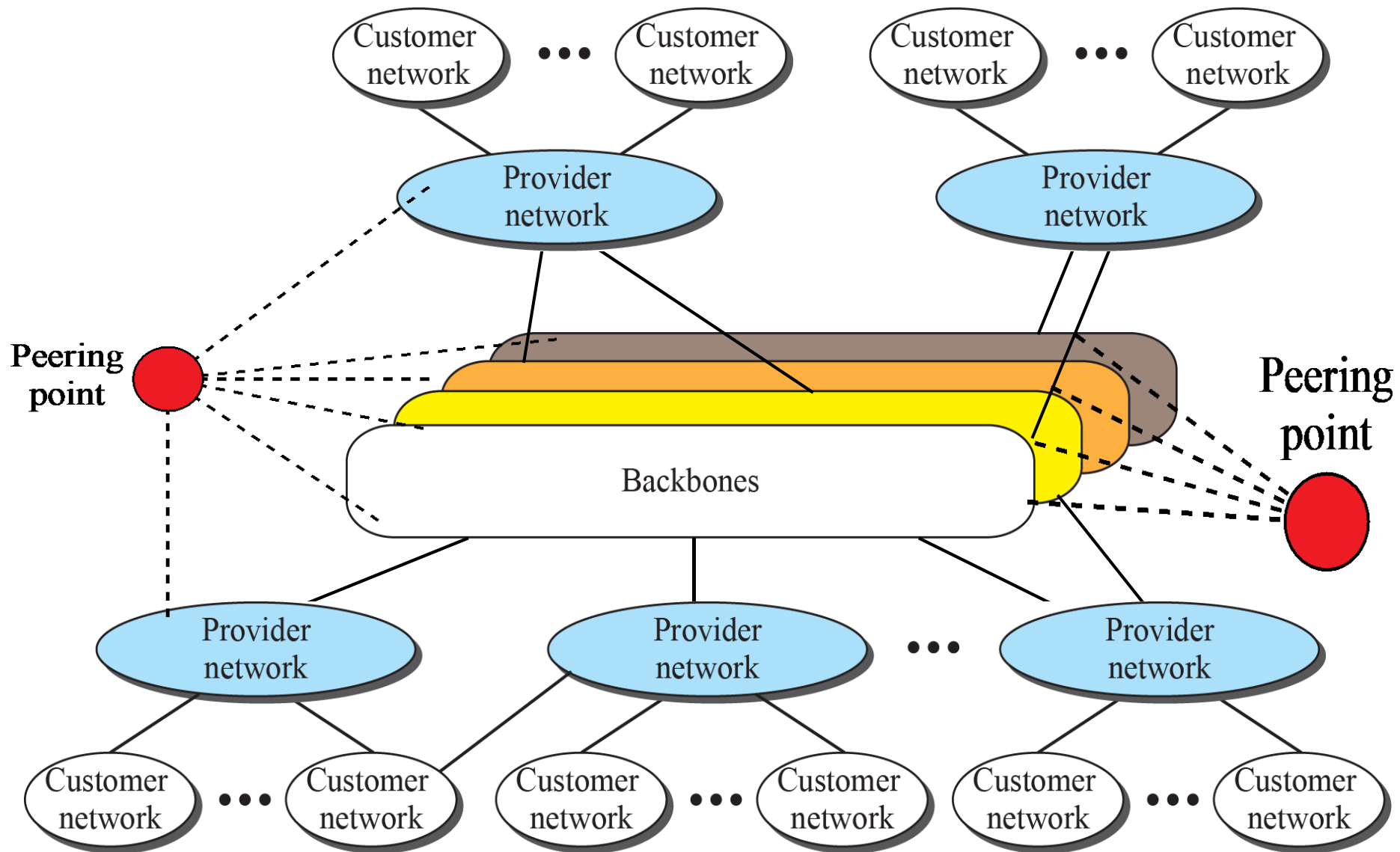
Internetwork

- **Computer Network** - group of connected communicating devices
- **Internetwork (internet)** - Two or more networks interconnected - Interconnection of LAN, MAN and WAN
- **Internet** (uppercase I) is a global network of networks

Internetwork



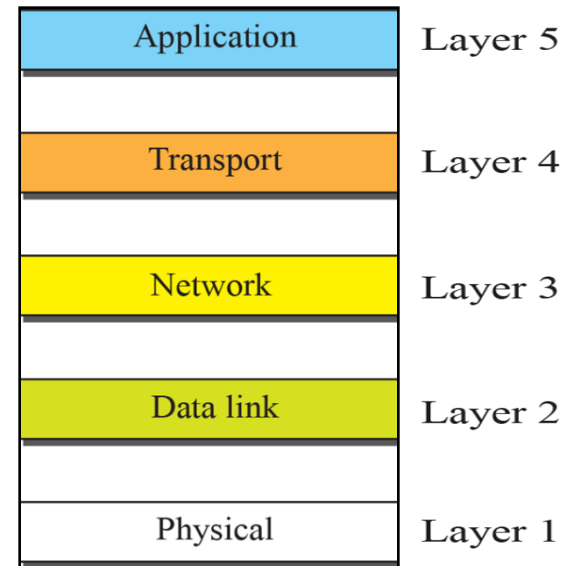
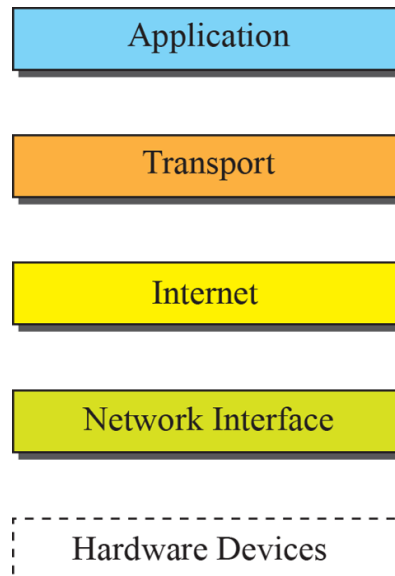
Internet



Network Models

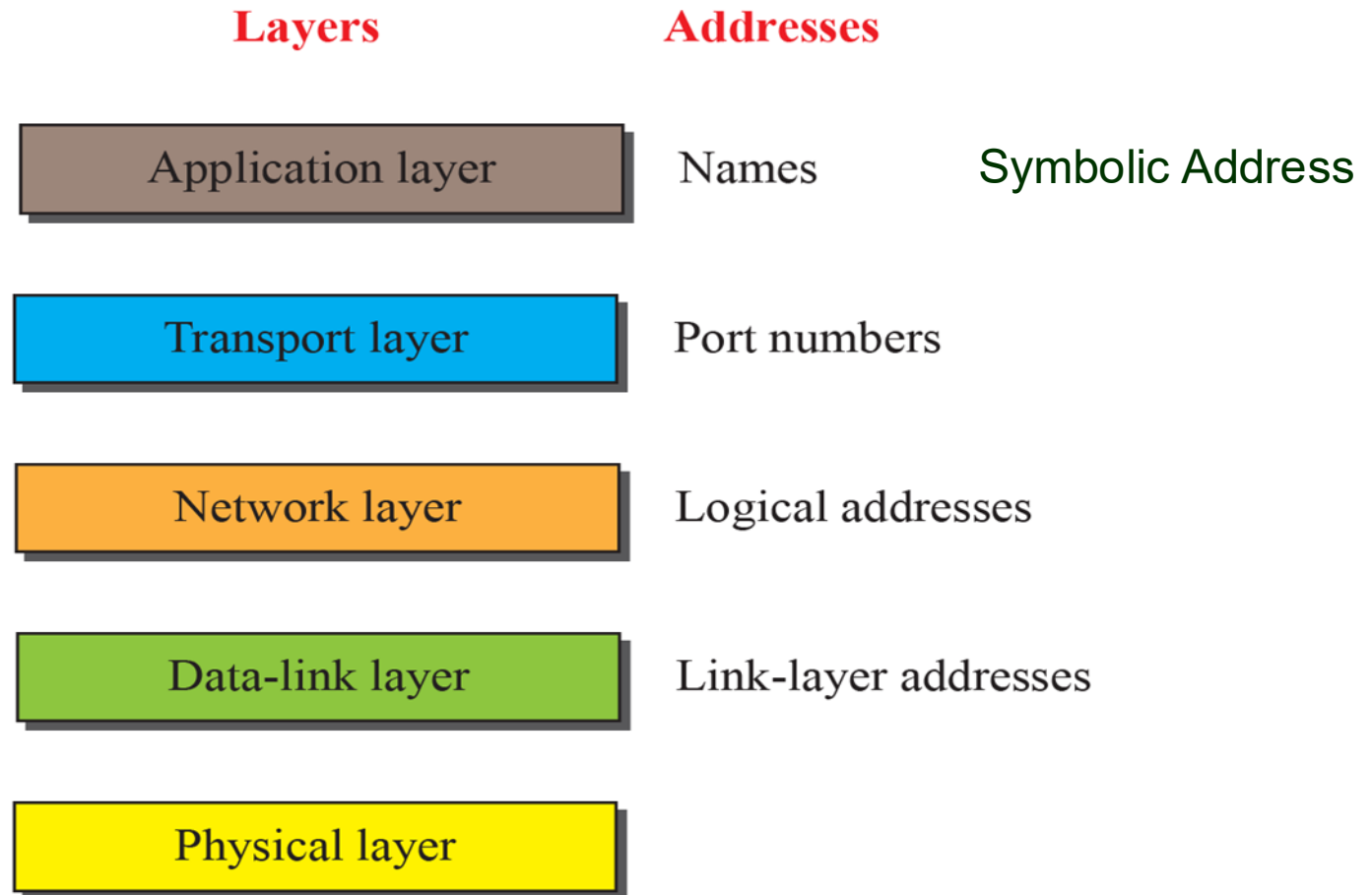
TCP/IP Protocol Suite

- Set of protocols organized in layers and used in the Internet today
- Originally thought as **4 layers** built **over the hardware**. But, **now treated as 5-layered**

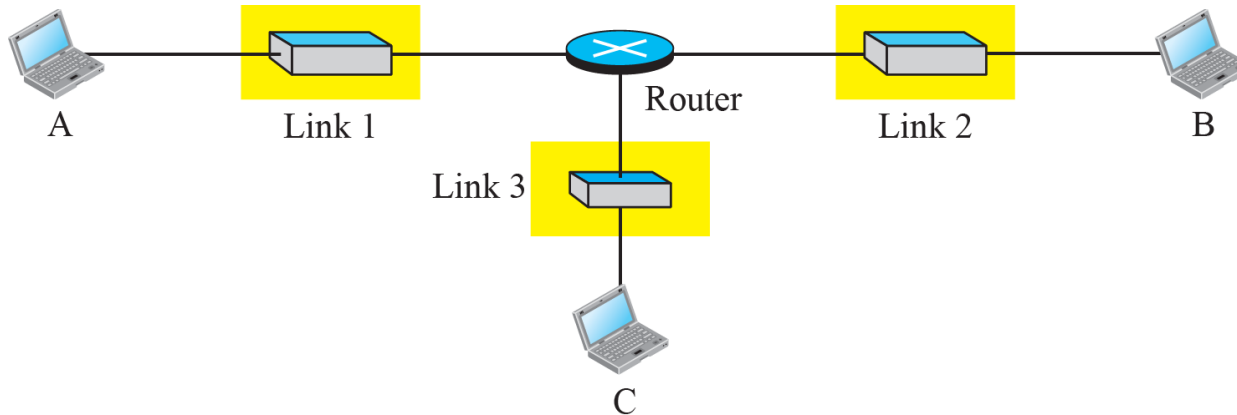


TCP/IP Addressing

- Addresses are used at 4 Layers
- At each layer – Source and Destination **Addresses**

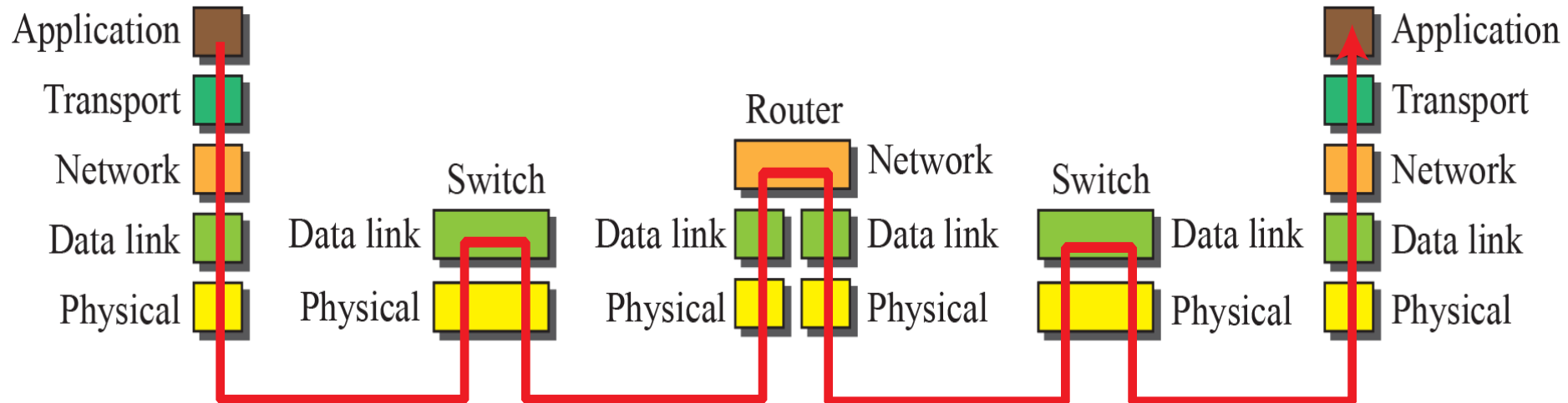


TCP/IP Protocol Suite



Source (A)

Destination (B)



TCP/IP Addressing

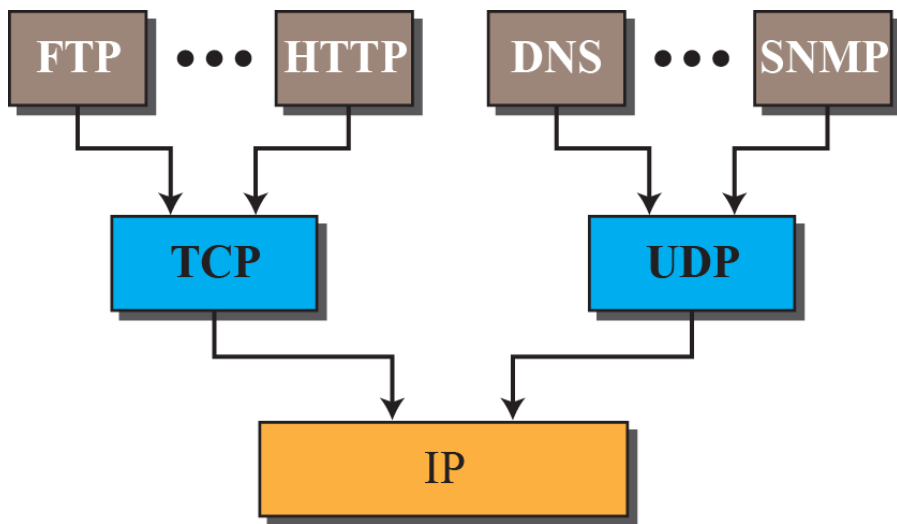
Addresses - used at 4 Layers

- **Application Layer: Name/ Symbolic Address**
E.g. `www.ibm.com`, `abc@gmail.com`
- **Transport Layer: Port Address** (Process IDs)
 - 16-bits (2-Bytes) - Decimal. E.g. `HTTP(80)`, `FTP(67)`...
- **Network Layer: Logical Address** (IP Addresses)
 - 32-bits (4-Bytes) - Dotted Decimal. E.g. `192.168.40.78`
- **Data Link Layer: Link Layer/ Physical/ MAC Address**
 - 48-bits (6-byte) – Hexadecimal. E.g. `07:01:02:01:2C:4B`

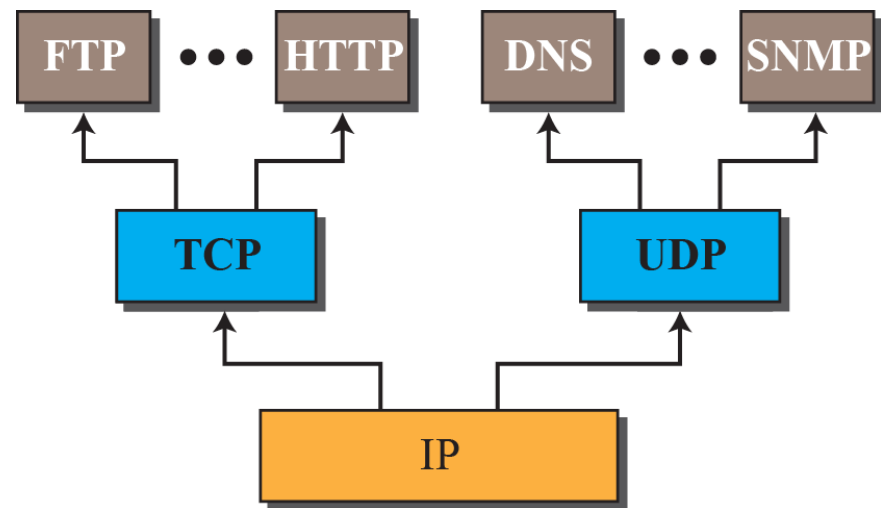
Multiplexing and Demultiplexing

Multiplexing: A protocol at a layer can encapsulate a packet from several next-higher layer protocols (one at a time)

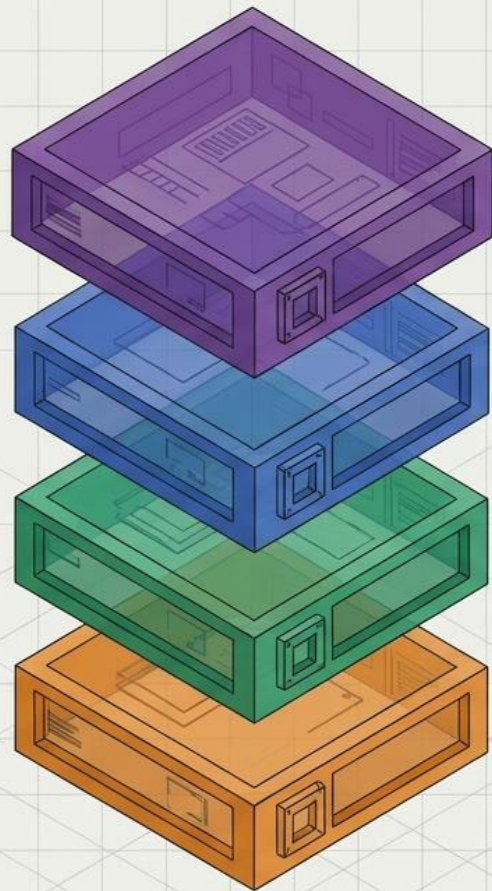
Demultiplexing: A protocol can decapsulate and deliver a packet to several next-higher layer protocols



a. Multiplexing at source



b. Demultiplexing at destination



Application Layer

Handles user-facing network applications and secure data formatting.

Transport Layer

Ensures reliable, error-free end-to-end data delivery.

Internet Layer

Manages addressing and routing data across distinct networks.

Network Access Layer

Controls physical hardware and localized signal transmission.

OSI Models

Physical Layer

Responsible for carrying individual bits across a link

Functions/ Responsibilities/ Services:

- Deals with the Mechanical and Electrical specifications of the interface and the transmission medium
- Defines the Procedures and Functions that the devices and interfaces have to perform
- **Design Issues:**
 - **Transmission Mode:** Simplex/ Half-duplex/ Duplex
 - **Physical Topology:** Mesh/ Star/ Ring/ Bus/ Hybrid
 - **Representation of bits:** Electrical/ Optical
 - **Encoding:** Signal format
 - **Signal Transmission**
 - **Data Rate:** Number of bits sent each second
 - **Media and Sharing:** Twisted Pair, Optical Fiber, Wireless

Data Link Layer

- Responsible for **node-to-node (hop-to-hop) delivery** of frames across direct connections

Functions/ Services/ Design Issues

- **Framing**: Division of stream of bits into **Frames**
- **Physical Addressing**: A unique 6 byte address imprinted in the **Network Interface Card (NIC)** - also called MAC address/ MAC Id/ Link Layer Address
- **Flow Control**: Handling speed mismatch between sender and receiver
- **Error Control**: Detect Errors in frames, Duplicate frames
- **Access Control**: When a link is shared, to determine which device can use the link at any given time (**Medium Access Control - MAC**)
- **Protocols**: **High Level Data Link Control (HDLC)**
Point-to-Point Protocol (PPP)

Network Layer

- Responsible for **Host-to-Host delivery** of packets - from the source host to the destination host

Functions/ Services/ Design Issues

- **Logical Addressing:** To identify every host uniquely
- **Routing:** Determine paths across the networks. The routers/ switches that interconnect different networks perform routing using routing tables: Protocols for Unicasting and Multicasting
- **Congestion Control:** To deal with traffic surges
- **Internetworking:** Handle incompatibilities between different networks through which the packets may move.

Main protocol: **Internet Protocol (IP)** - Connectionless protocol and **does not** provide Flow Control, Error Control, and Congestion Control

Auxiliary protocols - help IP in its delivery and routing tasks

Network Layer....

Auxiliary Protocols

- **Internet Control Message Protocol (ICMP)** helps to report some problems when routing a packet
- **Internet Group Management Protocol (IGMP)** helps in multicasting
- **Dynamic Host Configuration Protocol (DHCP)** helps to get the network-layer address for a host
- **Address Resolution Protocol (ARP)** helps to find the link-layer address of a host or a router when its network-layer address is given

Transport Layer

- **Process-to-Process delivery** of messages as - from a process in the source host to a process in the destination host

Functions/ Services/ Design Issues

- **Service Point Addressing/ Port addressing:** To identify the processes. Multiplexing of network layer connections – upward/ downward
- **Segmentation and Reassembly:** Dividing the message into transmittable segments and combining them at the destination
- **Connection Control:** Perform connection setup, maintenance, and release in connection oriented service. (Can provide connectionless service also)
- **Flow Control:** End-to-end flow control
- **Error Control:** Detect errors in messages

...Transport Layer

Three protocols: **TCP**, **UDP** and **SCTP**

1. **Transmission Control Protocol (TCP):** Connection oriented – Provides Flow Control, Error Control and Congestion Control
2. **User Datagram Protocol (UDP):** Connectionless - No Flow Control, Error Control and Congestion Control – useful for short messages
3. **Stream Controlled Transmission Protocol (SCTP):** Has the features of TCP and UDP

Application Layer...

Applications exchange messages using different protocols - many pre-defined protocols

- **Hypertext Transfer Protocol (HTTP)** – used for accessing the **World Wide Web (WWW)**
- **Simple Mail Transfer Protocol (SMTP)** - used for electronic mail (e-mail) service
- **File Transfer Protocol (FTP)** - used for transferring files from one host to another
- **Terminal Network (TELNET) and Secure Shell (SSH)** - used for accessing a site remotely

A university network administrator observes that webpages are loading correctly, but video conferencing applications frequently disconnect. Analyze which TCP/IP layer is most likely causing the problem and justify your answer.

Answer:

The issue is likely related to the **Transport Layer**.

- Video conferencing requires continuous transport of data.
- Packet loss, congestion, or transport protocol issues can affect video communication more severely than webpage loading.
- TCP retransmissions or UDP packet losses may cause interruptions.

...Application Layer

- **Simple Network Management Protocol (SNMP)** - used by an administrator to manage the Internet at global and local levels
- **Domain Name System (DNS)** - used by other protocols to find the network-layer address of a computer

Practice questions:

1. During campus recruitment, hundreds of students simultaneously access the university placement portal to register for companies, upload resumes, check interview schedules, and attend online aptitude tests. Explain how the TCP/IP protocol suite ensures successful communication between students and the placement server. Illustrate the role of each TCP/IP layer during this process.

2.

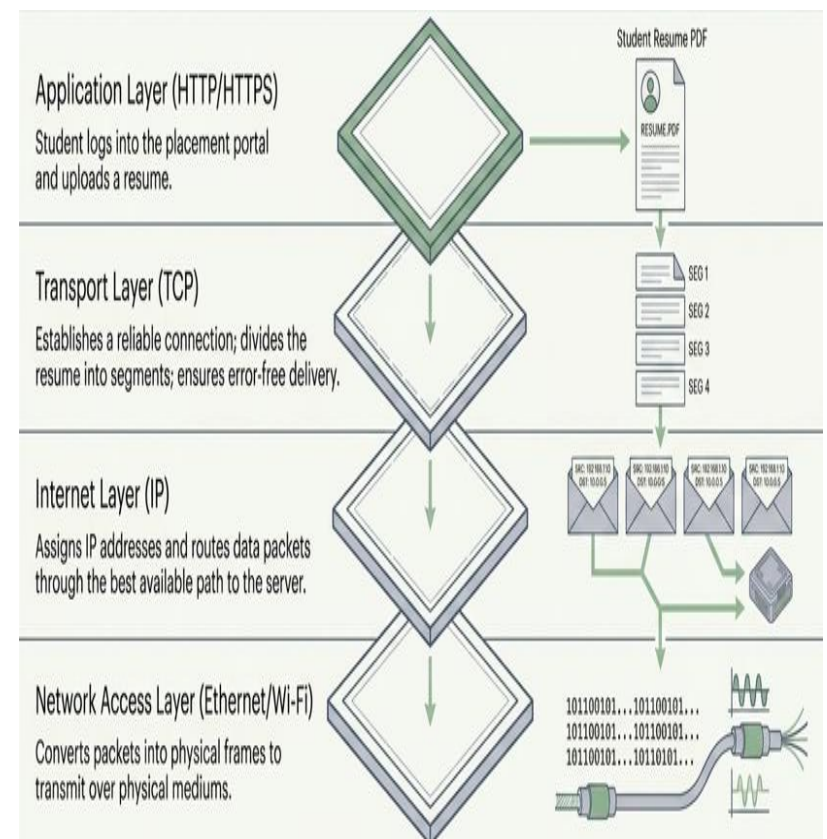
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Application Layer: Uses HTTP/HTTPS to allow students to register, upload resumes, check interview schedules, and attend online aptitude tests.

Transport Layer: Uses TCP to establish a reliable connection, divide data into segments, ensure error-free delivery, and retransmit lost packets.

Internet Layer: Uses IP addresses to route data packets from students' devices to the placement server through the best available path.

Network Access Layer: Transmits data over Wi-Fi or Ethernet by converting packets into frames and sending them through the physical network.



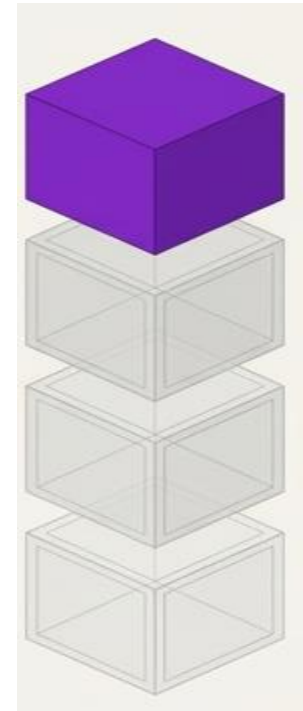
An online shopping website allows users to browse products, search items, and add them to the cart successfully. However, when users click "Pay Now," the payment page displays "503 Service Unavailable," while the rest of the website continues to work. Based on this scenario, identify the TCP/IP layer responsible, explain why only the payment process fails, identify the protocol used during payment, and state the first thing you would check to troubleshoot the issue.

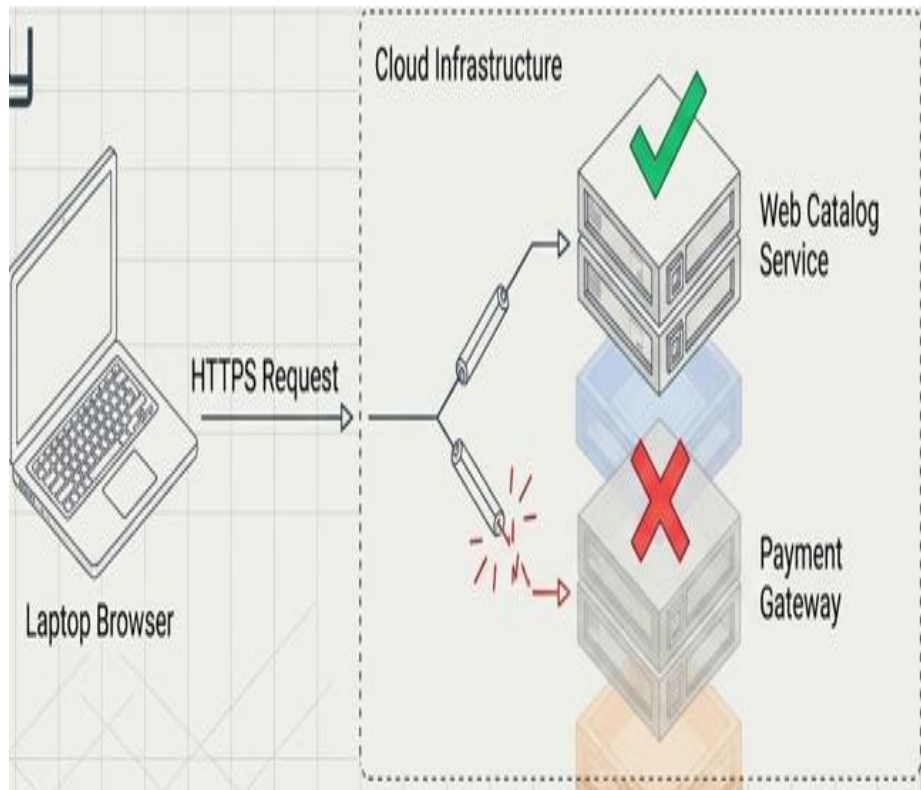
1. Which TCP/IP layer is responsible?

Answers:

Symptom: Core site functioning, checkout fails.

Diagnosis: Application Layer Issue.





Protocol

HTTPS (Hypertext Transfer Protocol Secure). Protects sensitive payment data between browser and server.

The Mechanism

Modern websites distribute functions. Browsing works because the catalog service is online; payment fails because the distinct payment gateway service is unreachable or overloaded.

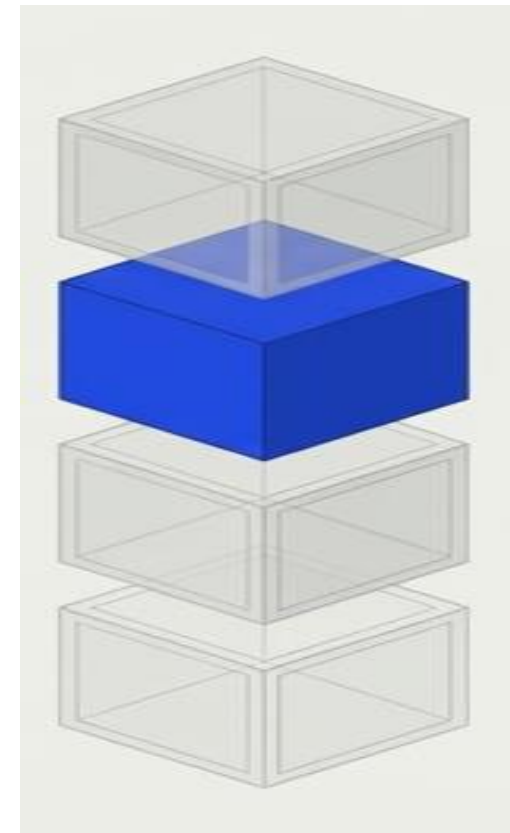
First Response

Check if the payment server is reachable, verify the HTTPS service is functioning, and review application/server logs for crash errors.

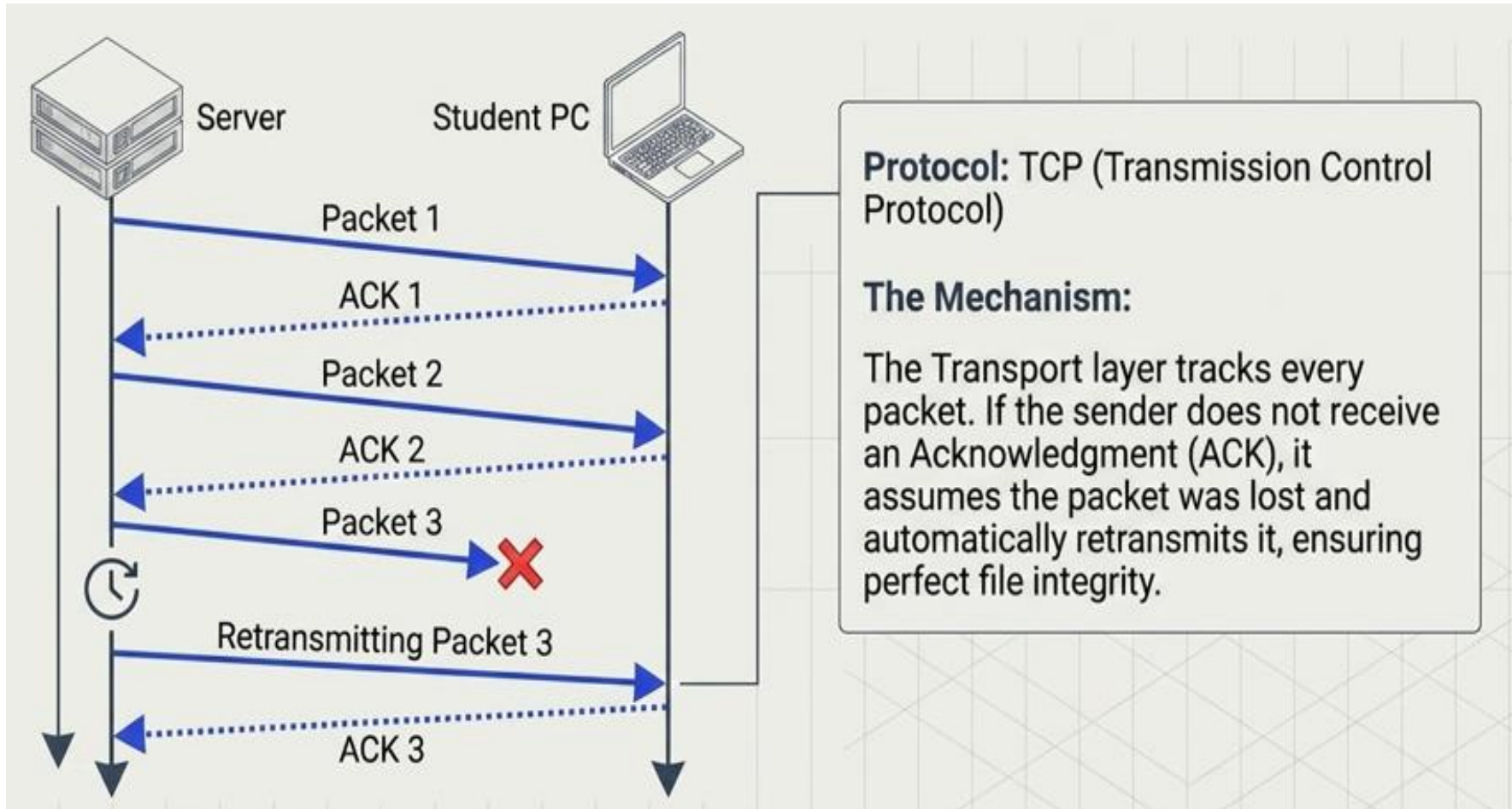
A student is downloading a 2 GB software update. During the download, the internet connection becomes unstable, and a few data packets are lost. However, after some time, the download completes successfully without any corrupted files. Based on this scenario, identify the TCP/IP layer responsible, explain how the file was downloaded correctly despite packet loss, identify the protocol used, and state the mechanism that ensures reliable delivery.

Symptom: Lost packets during large transfer.

Diagnosis: Transport Layer Rescue.



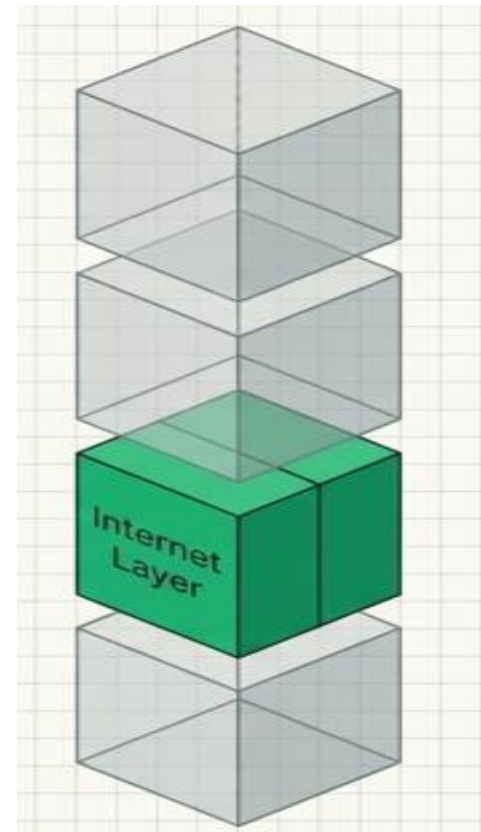
TCP Packet Recovery Sequence



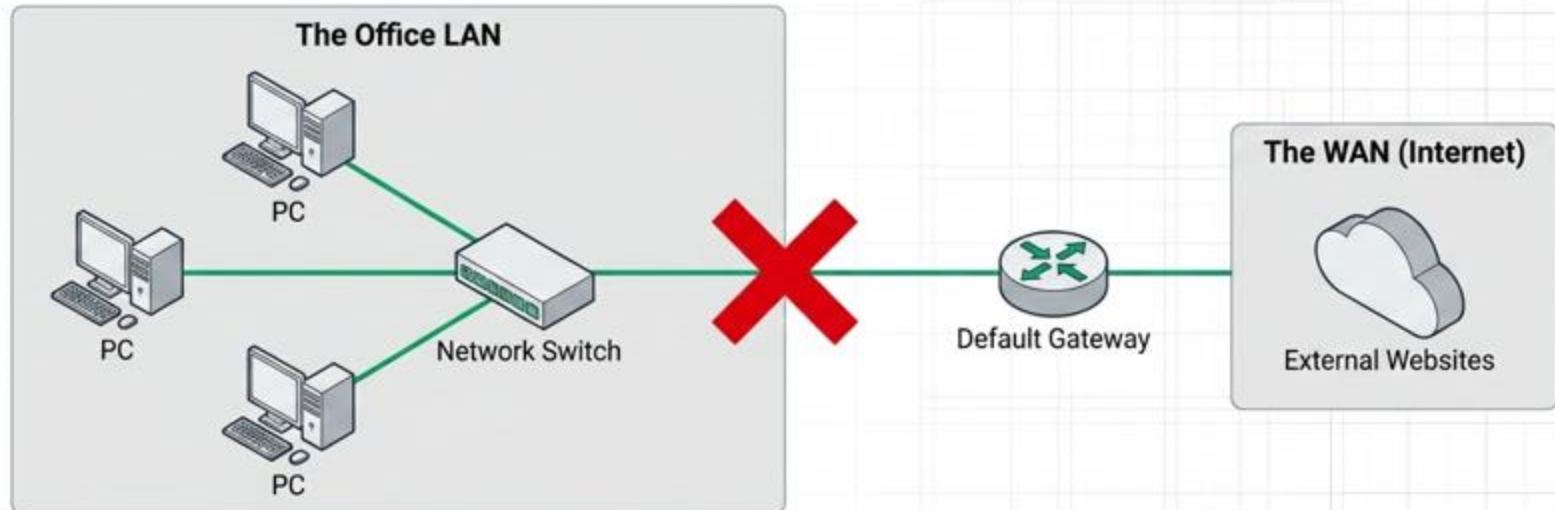
An employee in a company can communicate with other computers on the same office network but cannot access websites or servers on the Internet. The network cable is connected properly, and the computer has a valid IP address. Based on this scenario, identify the TCP/IP layer responsible, explain the possible cause of the problem, identify the protocol involved, and state the first troubleshooting step.

Symptom: Local access only, external blackout.

Diagnosis: Internet Layer Issue.



Boundary Failure: Trapped on the Local Network



Protocol

IP (Internet Protocol).

The Mechanism

The Internet layer manages routing data between different networks. Local traffic doesn't need routing, but external traffic requires a functional gateway to exit the local subnet.

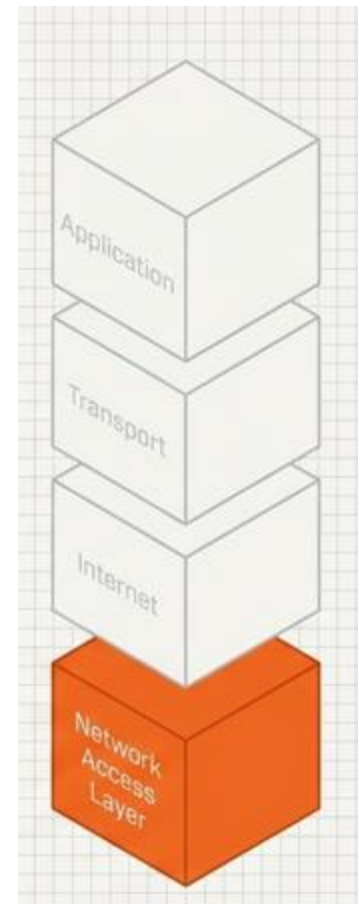
First Response

Verify the default gateway IP configuration and check the router's status.

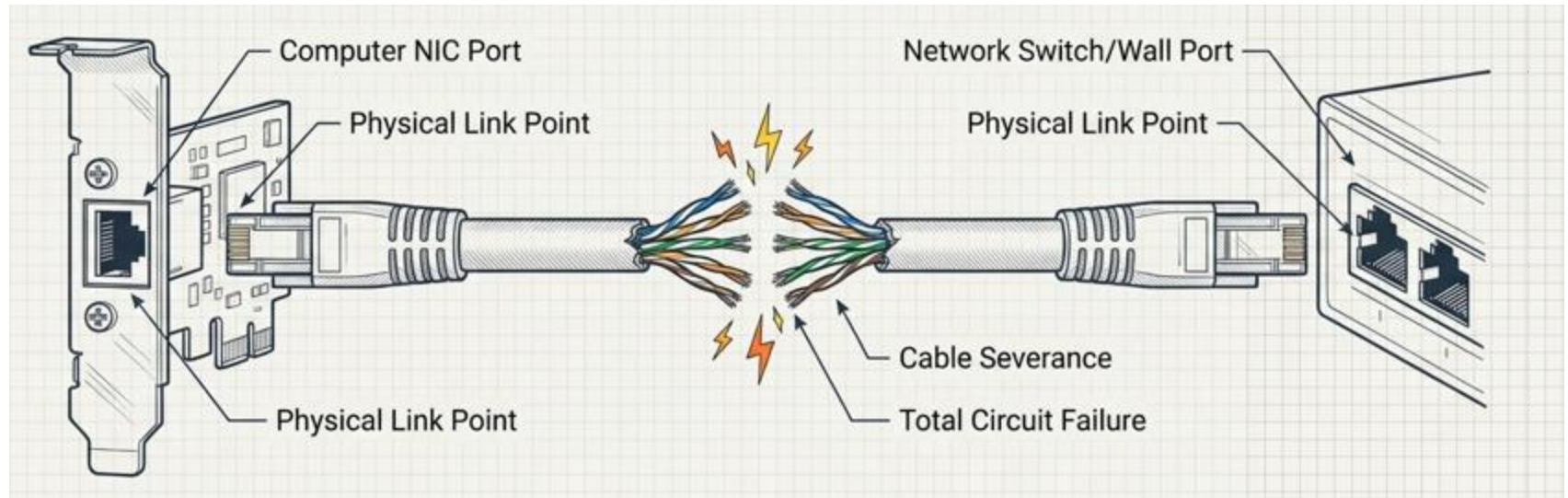
A desktop computer suddenly loses its network connection after the Ethernet cable is accidentally damaged. The computer cannot communicate with any device on the network, even though its IP address is correctly configured. Based on this scenario, identify the TCP/IP layer responsible, explain why communication fails, identify the hardware components involved, and state the first troubleshooting step.

Symptom: Complete isolation despite valid software configuration.

Diagnosis: Network Access Layer Issue.



Physical Layer Failure: The Broken Connection



Hardware Involved

Ethernet Cable, Network Interface Card (NIC), and Network Switch.

The Mechanism

Software configurations (like valid IP addresses) cannot function if the physical medium required to transmit electrical or light signals is severed.

First Response

Physically inspect and replace the damaged Ethernet cable; check the switch port for link lights.

Physical Layer and Media

Data and Signals

- Data must be converted into signals for transmission.
- Data and the signals that represent them can be in **analog** or **digital** form:
 - a)Analog data: continuous values (infinite values within a range).**
 - b)Digital data: discrete values (limited number of values).**
- Signals can be periodic or aperiodic (non-periodic):
 - a)Periodic signals: have repeating patterns (cycles) in identical periods.**
 - b)Aperiodic signals: have no regular patterns..**

Periodic Analog Signals

Characteristics of Periodic Signals

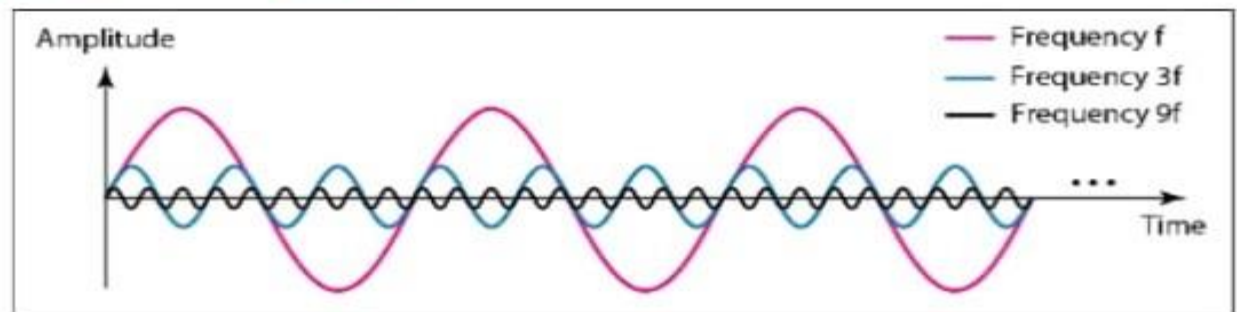
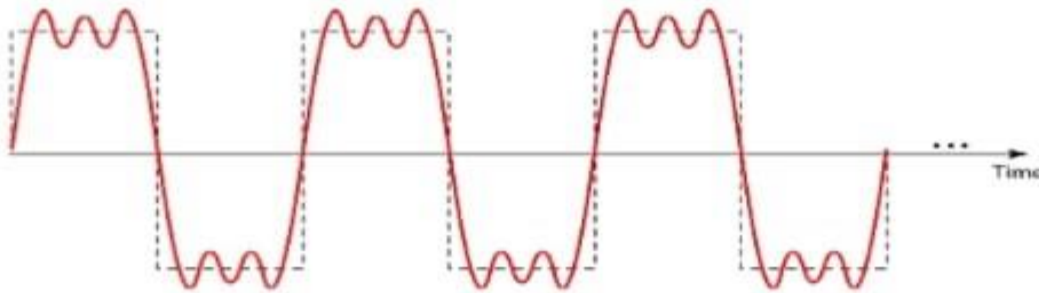
- **Peak Amplitude:** Highest intensity, proportional to the energy.
Eg. Voltage (volts) for electrical signals
-- Expressed as Absolute value or RMS value
- **Frequency (f) :** Number of cycles/ second: (Hertz - Hz)
Time in seconds to complete 1 cycle - Period (T)
 $f = 1/T$
- **Phase:** Describes the position of the waveform relative to time 0 (phase shift) – Measured in degrees/ radians

Periodic Analog Signals

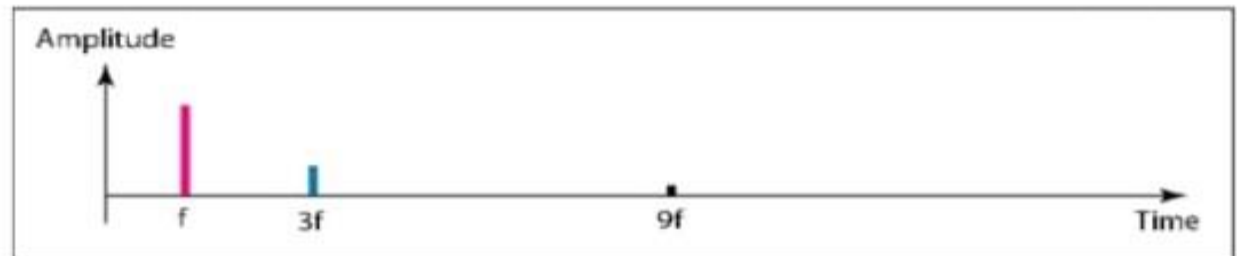
Composite Signals

- In order to communicate data, composite signals are required
- Fourier has shown that:
 - a) a periodic composite signal can be decomposed into finite number of sine waves with integer valued frequencies**
 - b) a non-periodic signal can be decomposed into infinite sine waves (harmonics) with real valued (continuous) frequencies**
- Frequency Of D Component signal which is the same as the frequency of Composite signal is called the fundamental frequency or the first harmonic

A Composite Periodic Signal

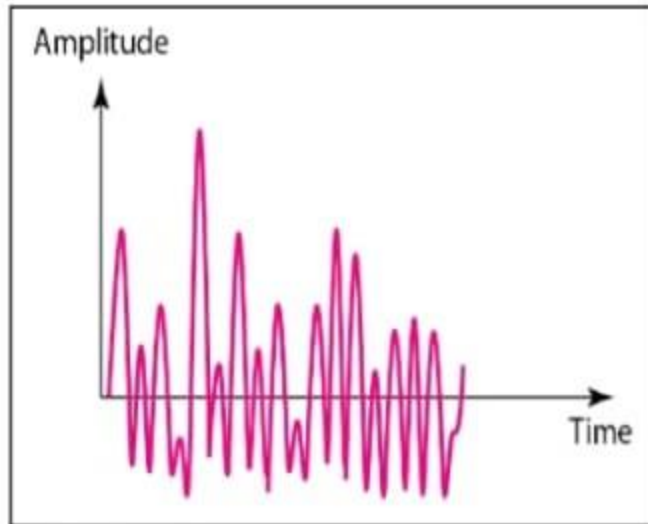


a. Time-domain decomposition of a composite signal

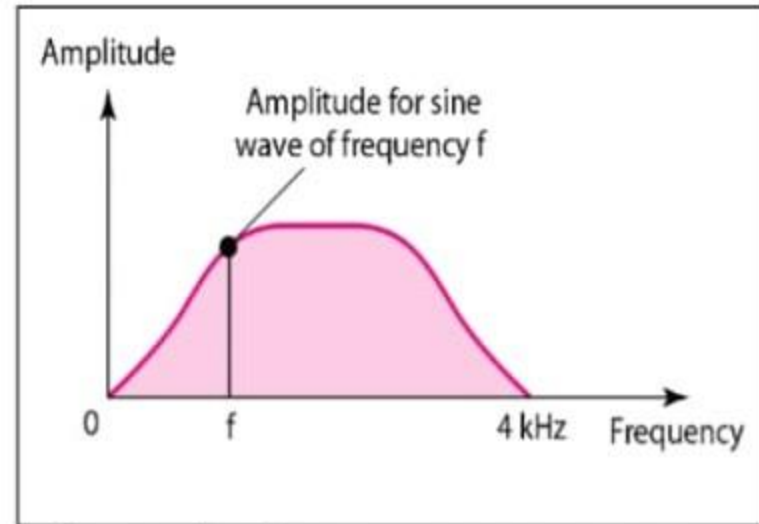


b. Frequency-domain decomposition of the composite signal

A Composite Non-periodic Signal



a. Time domain



b. Frequency domain

Periodic Analog Signals

Bandwidth

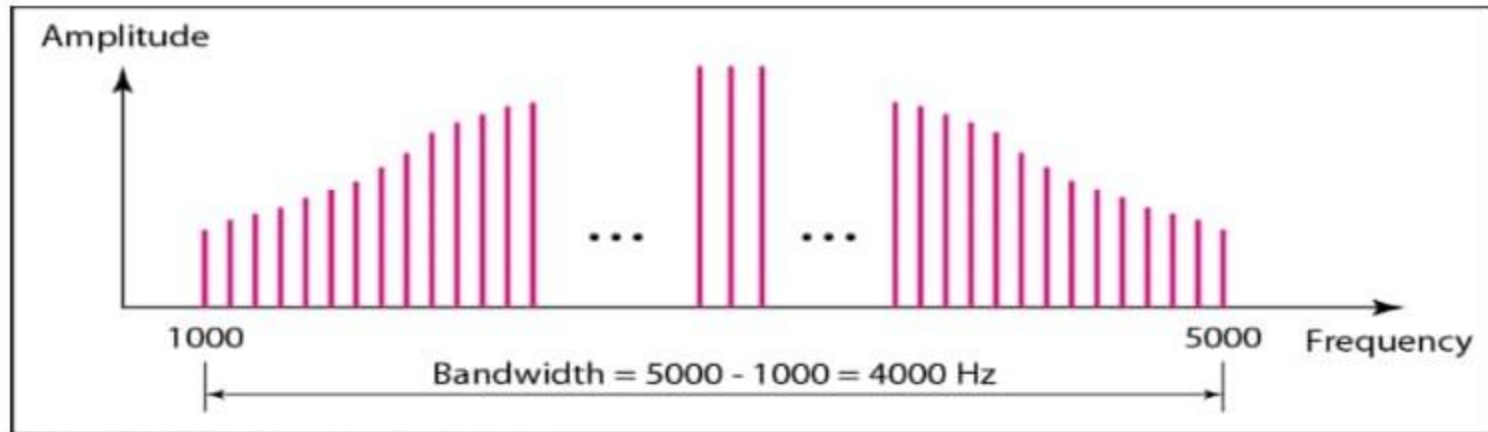
Difference between the highest and the lowest frequencies contained in a composite signal – bandwidth

Formula:

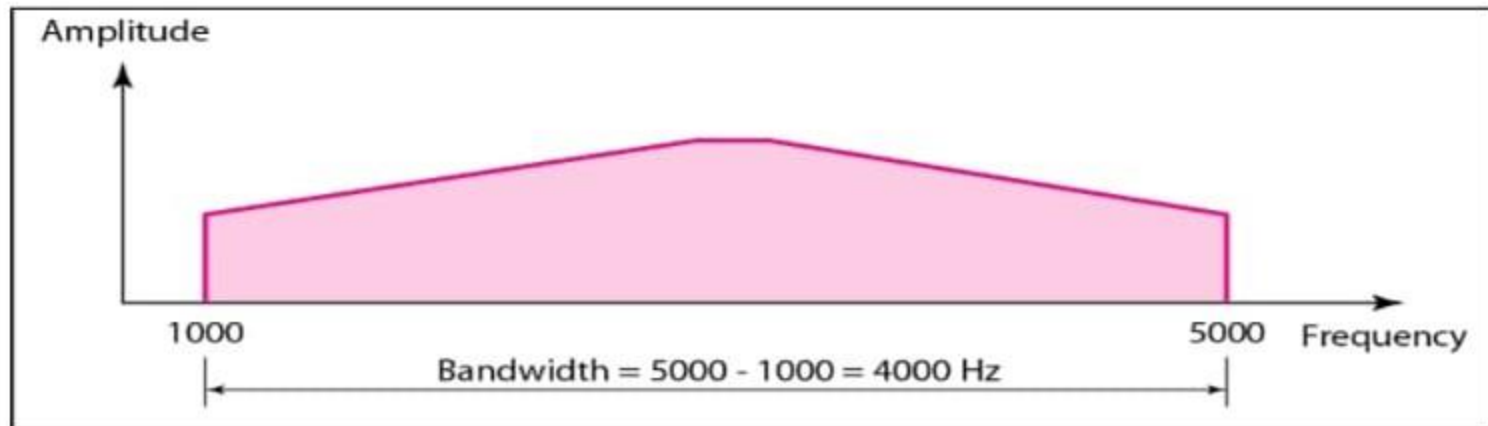
$$B = f_h - f_l$$

Human voice - non-periodic composite signal with continuous frequencies in the range 0–4 KHz

Bandwidth



a. Bandwidth of a periodic signal



b. Bandwidth of a nonperiodic signal

Digital Signals

- Digital signals are described by the bit rate – which is the number of bits transmitted per second (bps).
- Information can also be represented by a digital signal.
 - a) If a digital signal has 2 levels, we can send 1 bit per level (0 or 1).**
 - b) If a digital signal has 4 levels, we can send 2 bits per level (00, 01, 10, or 11).**
 - c) If a digital signal has m levels, we can send $\log_2 m$ bits per level. In other words, to transmit k bits per level, 2^k levels are required.**
- Bit length is the distance that one bit occupies on the medium.
Bit length = Propagation speed × Bit duration

Digital Signals

1) It is required to download text documents at the rate of 100 pages per minute. What is the required bit rate of the channel?

$$(100 \times 2400 \times 8) / 60 = 25600 \text{ bps} = 25.6 \text{ kbps}$$

2) A digitized voice channel is made by digitizing a 4-kHz band-limited analog voice signal. We need to sample the signal at twice the bandwidth. If each sample has 8 bits. What is the required bit rate?

$$2 \times 4000 \times 8 = 64000 \text{ bps} = 64 \text{ kbps}$$

Digital signal as composite analog signal

- A digital signal is a composite analog signal with an infinite bandwidth
 - If the digital signal is **periodic** (rarely it is so) the decomposed signal has infinite bandwidth with discrete frequencies
 - If the digital signal is **non-periodic** the decomposed signal has infinite bandwidth with continuous frequencies



a. Time and frequency domains of periodic digital signal



b. Time and frequency domains of nonperiodic digital signal

Transmission Impairments

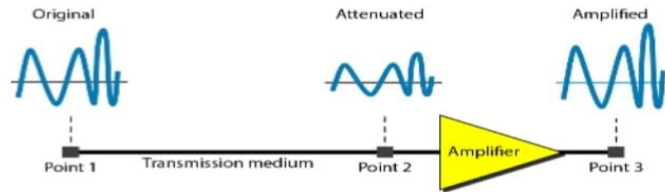
Media are not perfect: imperfections in media cause signal impairment, meaning the signal at the beginning of the medium is not the same as the signal at the end of the medium.

Three causes of impairment are **attenuation, distortion, and noise**.

- **Attenuation** is the **reduction in the strength** of the signal.
- **Distortion** is the **change in the shape** of the signal.
- **Noise** is an **extraneous signal** that corrupts the signal.

Transmission Impairments – Attenuation

- Attenuation is the reduction in the strength of the signal, it is the loss of energy.
- Amplifiers are required to compensate for the loss.



- If P_1 and P_2 are the powers of a signal at points 1 and 2, their relative strength is measured in decibels, defined as a ratio:
$$\text{dB} = 10 \log_{10}(P_2 / P_1)$$
- Decibel is –ve if the signal is **attenuated** ($P_2 < P_1$) or +ve if the signal is amplified ($P_2 > P_1$).
- Instead of power, if voltage is available:

$$\text{dB} = 20 \log_{10}(V_2 / V_1)$$

Transmission Impairments – Attenuation

- Suppose the signal power is reduced to one-half, i.e., $P_2 = 0.5P_1$, the attenuation is:

$$10 \log_{10}(P_2 / P_1) = 10 \log_{10}(0.5P_1 / P_1) = 10 \log_{10}(0.5) = 10(-0.3) = -3 \text{ dB}$$

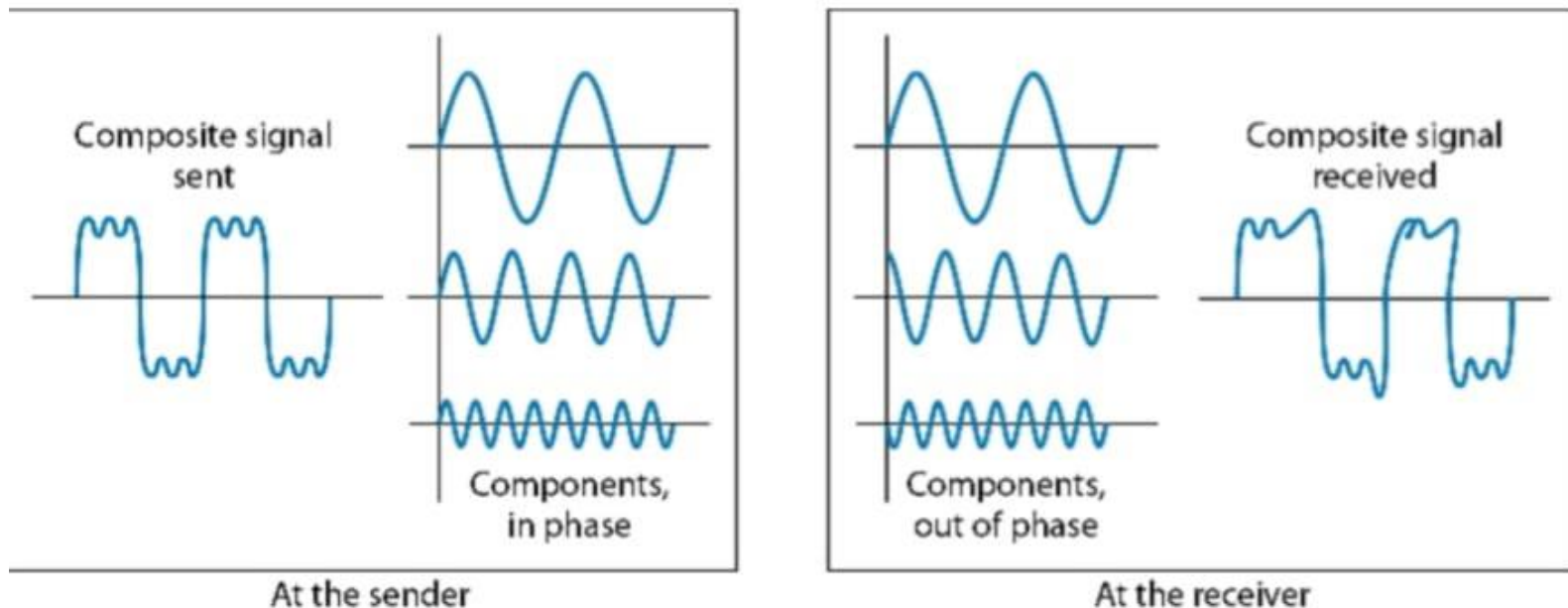
Hence, a loss of 3 dB (–3 dB) is equivalent to losing one-half of the power.

- Suppose the signal power is increased by 10 times, i.e., $P_2 = 10P_1$, the amplification is:

$$10 \log_{10}(P_2 / P_1) = 10 \log_{10}(10P_1 / P_1) = 10 \log_{10}(10) = 10(1) = 10 \text{ dB}$$

Transmission Impairments – Distortion

- Distortion is the change in the shape of the signal.
- Each component frequency of a composite signal has its own speed in the transmission medium.
- Results in a difference in the phases of different components and hence the change in the shape

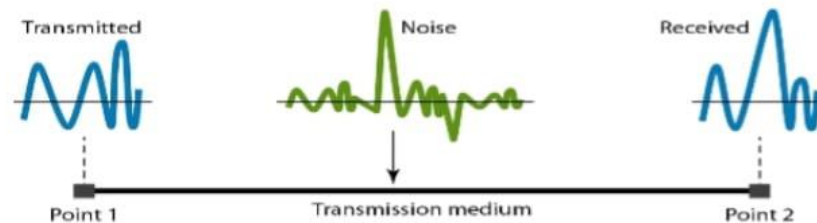


Transmission Impairments – Noise

- Noise is an extraneous signal that corrupts the signal.

Types of Noise:

1. **Thermal Noise**: Due to the random motion of electrons in the wire.
2. **Induced Noise**: Due to motors and appliances.
3. **Impulse Noise**: Spike/surge due to power lines, lightning, etc.
4. **Cross-Talk**: Due to adjacent transmission lines. One line becomes an antenna, and the adjacent becomes a receiver.



Data Rate Limits

Data rate is the speed at which the data is transmitted — depends on three factors:

1. Bandwidth available
2. Level of the signals
3. Level of noise (quality of the channel)

Two formulae to calculate the data rate:

1. By **Nyquist** – For noiseless channels
2. By **Claude Shannon** – For noisy channels

Nyquist Data Rate / Nyquist Bit Rate

The maximum data rate for a noiseless channel is:

$$\text{Bit Rate} = 2 \times \text{BW} \times \log_2 L$$

BW – Bandwidth of the channel in Hz

L – Number of signal levels used

NOTE: Increasing the levels of a signal may reduce the reliability of the system

E.g. The maximum bit of a noiseless channel with a bandwidth of 3 kHz transmitting a signal with 2 signal levels is:

$$= 2 \times 3000 \times \log_2 2 = 6000 \text{ bps} = 6 \text{ kbps}$$

If 4 level signal is used the bit rate is:

$$= 2 \times 3000 \times \log_2 4 = 12000 \text{ bps} = 12 \text{ kbps}$$

Shannons Channel Capacity Theorem

The maximum data rate for a noisy channel is:

$$C = B \times \log_2(1 + \text{SNR}) \text{ bps}$$

B – Bandwidth of the channel in Hz

SNR – Signal to Noise Ratio

NOTE:1. The upper limit on the data rate is not dependent on the number of signal levels

2. $\log_2 X = (\log_{10} X) / (\log_{10} 2) = (\log_{10} X) / 0.3010$

Eg. A telephone line normally has a bandwidth of 3 KHz.

The Signal-to-Noise Ratio is usually 3162. For this channel the capacity is calculated as:

$$\begin{aligned} C &= 3000 \times \log_2(1 + 3162) \\ &= 3000 \times (\log_{10}(3163) / 0.3010) \\ &= 34884 \text{ bps} \end{aligned}$$

Performance

Described by:

1. Bandwidth
2. Throughput
3. Latency (Delay)
4. Bandwidth-Delay Product
5. Jitter

1. Bandwidth

Capacity of a channel to transmit data. In networking, the term bandwidth is used in two contexts:

- **Bandwidth in Hertz** – Range of frequencies in a composite signal or the range of frequencies that a channel can pass.
- **Bandwidth in bits per second** – Speed of bit transmission in a channel.

Performance

2. Throughput

Actual rate at which data is transmitted – measured in bps.

Example: A channel may have a bandwidth of 1 Mbps, but at a given time the channel may be transmitting 200 Kbps.

3. Latency (Delay)

Time taken by a message to arrive entirely at the destination from the instant the first bit is sent out from the source.

$$\text{Latency} = \text{Transmission time} + \text{Propagation time} + \text{Queuing time} + \text{Processing time}$$

Performance

4. Bandwidth-Delay Product

- The product of bandwidth and delay is the number of bits that can fill the link.

5. Jitter

- Non-uniformity in the delays for different parts of a message is called jitter.
- Jitter causes interruptions / breaks in real-time communication.

Digital Transmission

- **Data** (information) must be converted into **digital** or **analog signal** for transmitting
- Conversion of digital data into digital signals is termed **Digital-to-Digital Conversion**
- Conversion of analog data into digital signals is termed **Analog-to-Digital Conversion**

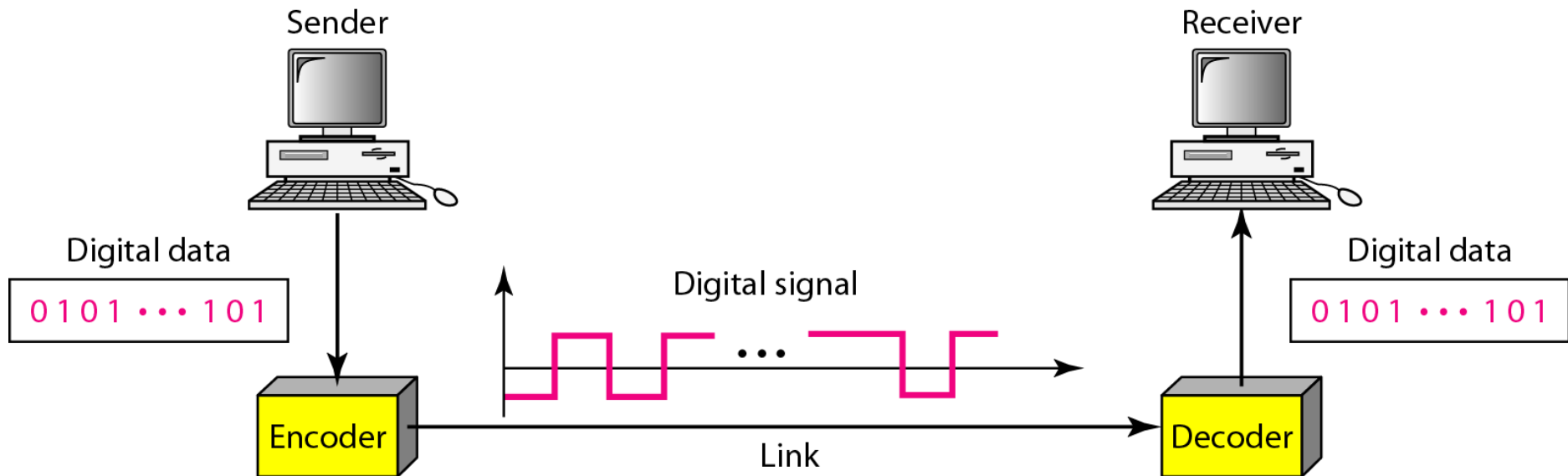
Digital-to-Digital Conversion can be done using two techniques:

1. **Line Coding**
2. **Block Coding**

Quality of the signal may be improved by **Scrambling**

Line Coding

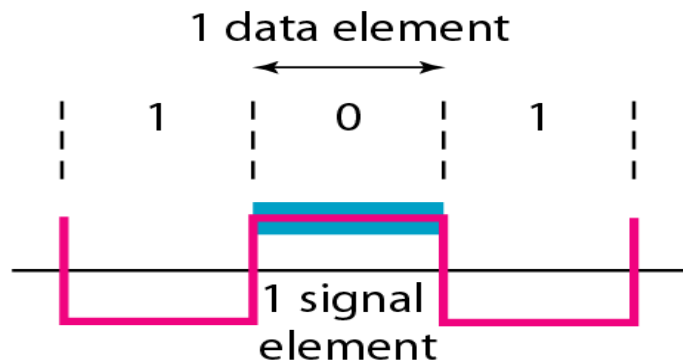
- **Line Coding** is the process of conversion of digital data into digital signals
- At the sender, digital data are converted into digital signal by **encoding**. At the receiver digital data are recreated using **decoding**



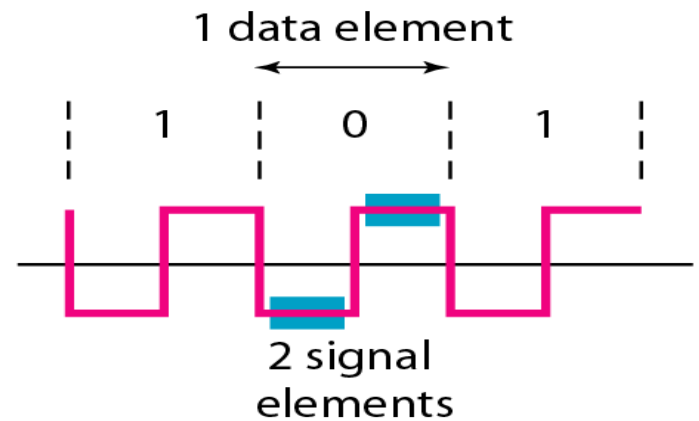
Line Coding

- In **Line Coding**, a bit or a group of bits is represented by one pulse (signal element) or a group of pulses
 - In other words **one pulse** may represent a part of a bit, or one bit or even more bits
- The **ratio** of number of bits to the number of pulses that represent them is denoted as **r**
 - r may be 1, $\frac{1}{2}$, 2, $\frac{4}{3}$
- The rate at which the **signal elements** are transmitted is called the **pulse rate** or **baud rate**
 - **If one pulse represents one bit, then, baud rate = bit rate**

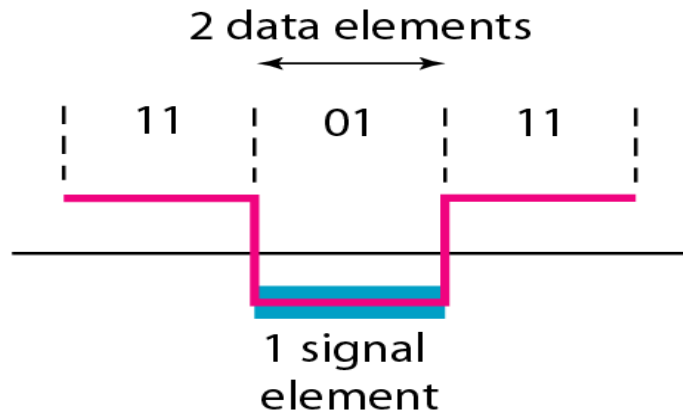
Line Coding



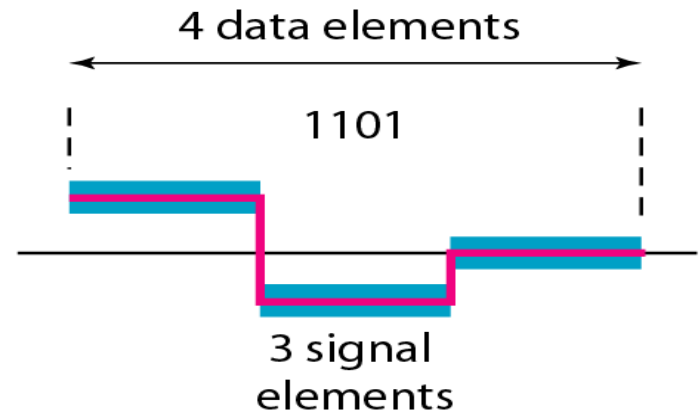
a. One data element per one signal element ($r = 1$)



b. One data element per two signal elements ($r = \frac{1}{2}$)



c. Two data elements per one signal element ($r = 2$)



d. Four data elements per three signal elements ($r = \frac{4}{3}$)

Line Coding

- One **goal** in data communication is to **increase the data rate** (increases the speed of transmission) and **decrease the signal rate** (reduces the bandwidth requirement)
- The best case is when the signal rate is minimized and the worst case is when the signal rate is maximized. This **case factor** can be in the range **0-1**
- **Signal rate** and **Data rate** can be related by a formula

$$S = c \times N \times (1/r)$$

S - Number of Signal elements - Baud

c - Case factor (between 0 and 1)

N - data rate (bps)

r - number of bits to number of pulses ratio

Line Coding

E.g.: A signal is carrying data where one data element is encoded as one signal element. If the bit rate is 100 kbps, and c is between 0 and 1, what is the average value of the **baud rate** ?

We have $r=1$ and $N=100$ kbps

Let $c=0.5$

$$\begin{aligned} S &= c \times N \times (1/r) \\ &= 0.5 \times 100,000 \times (1/1) \\ &= 50,000 \\ &= \mathbf{50 \text{ kbaud}} \end{aligned}$$

Line Coding

Since bandwidth is proportional to the signal rate, the minimum bandwidth is:

$$B_{\min} = c \times N \times (1/r)$$

c - Case factor (between 0 and 1)

N - data rate (bps)

r - number of bits to number of pulses ratio

Or

$$N_{\max} = (1/c) \times B \times r$$

Signal with **L** levels carries $\log_2 L$ bits per level. If the case factor **c** is chosen as (1/2), then,

$$N_{\max} = 2 \times B \times \log_2 L \quad \dots \dots \dots \text{Nyquist criterion}$$

Why Line Coding? - Problems

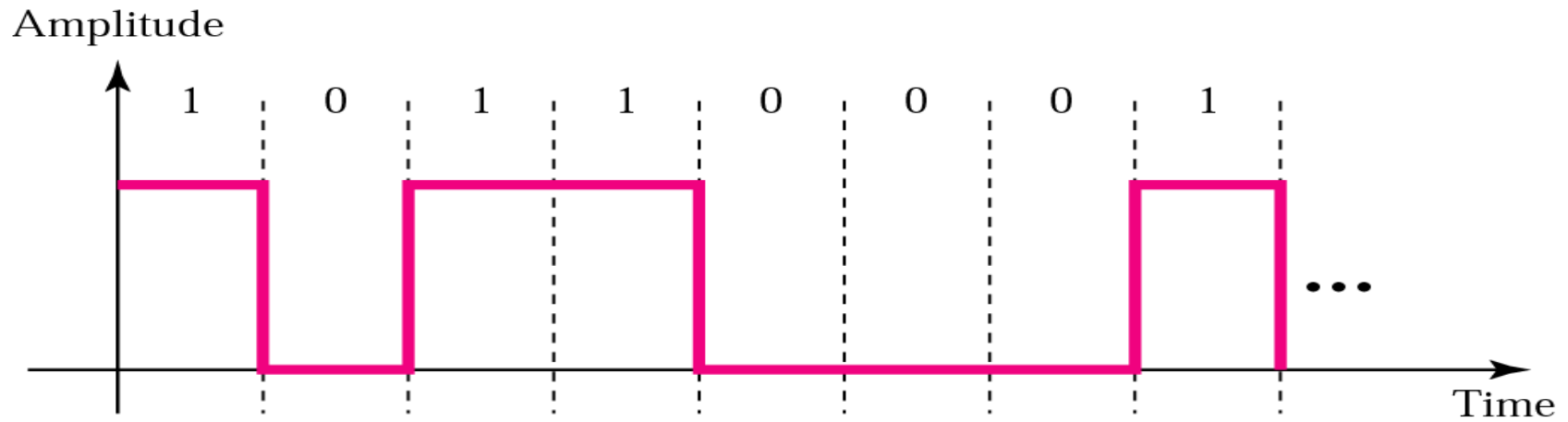
1. **Baseline Wandering:** A long string of 0's or 1's causes a drift in the baseline (average received signal power) – makes decoding difficult
2. **DC Components:** A long string of 0's or 1's is like DC that has very low frequency components - cannot pass through transmission line or components like transformers
3. **Synchronization:** If the sender and receivers clocks are not synchronized, the receiver might misinterpret the signals. Hence it is advisable to have a signal that has timing information also with it - **self-synchronising** signals

Why Line Coding? - Desirable features

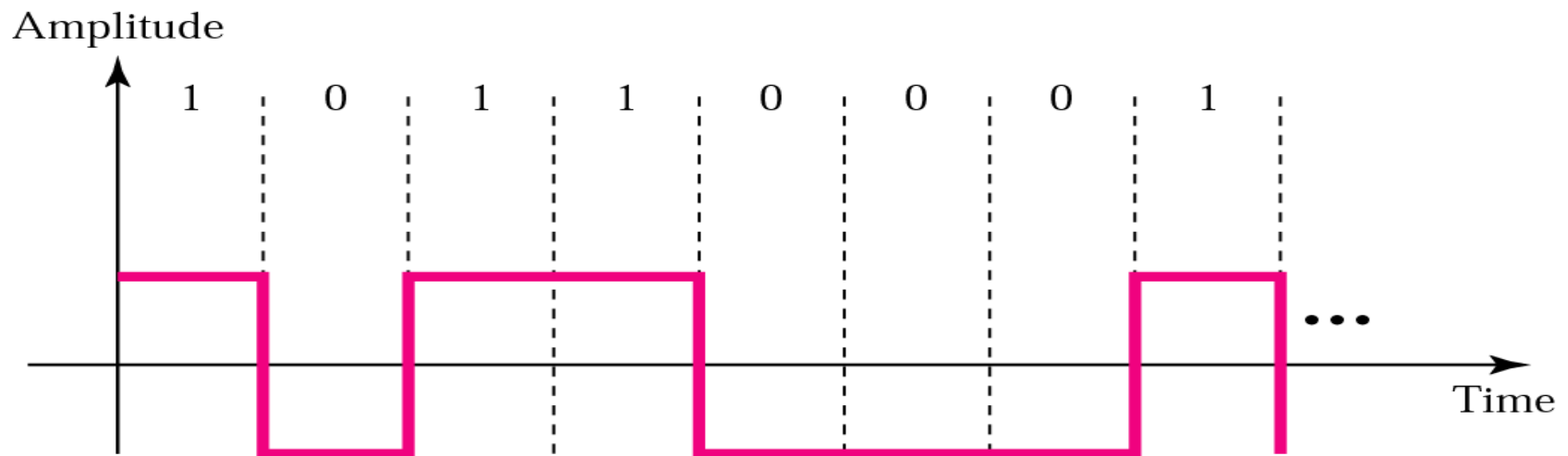
4. **Error detecting ability:** It is desirable to have built-in error-detecting capability in the signal
5. **Immunity:** It is desirable to have a signal that is immune to noise and interferences
6. **Easier scheme:** A signal with less number of levels is easy to interpret and easy to implement

The **problems** and **desirable features** have resulted in the development of schemes for encoding the bit stream into digital signals - **Line Coding** schemes

DC Components

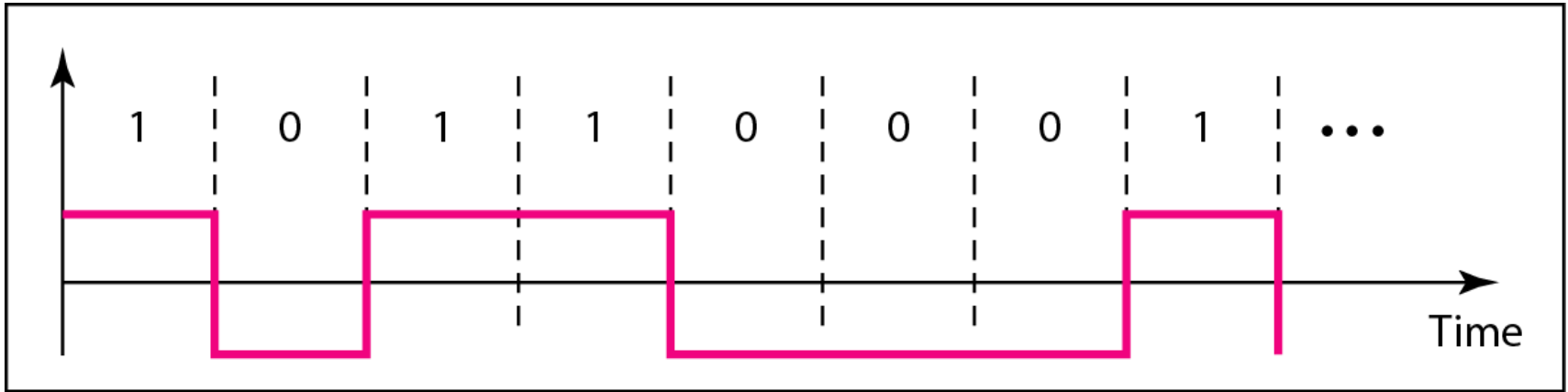


a. A signal with dc component

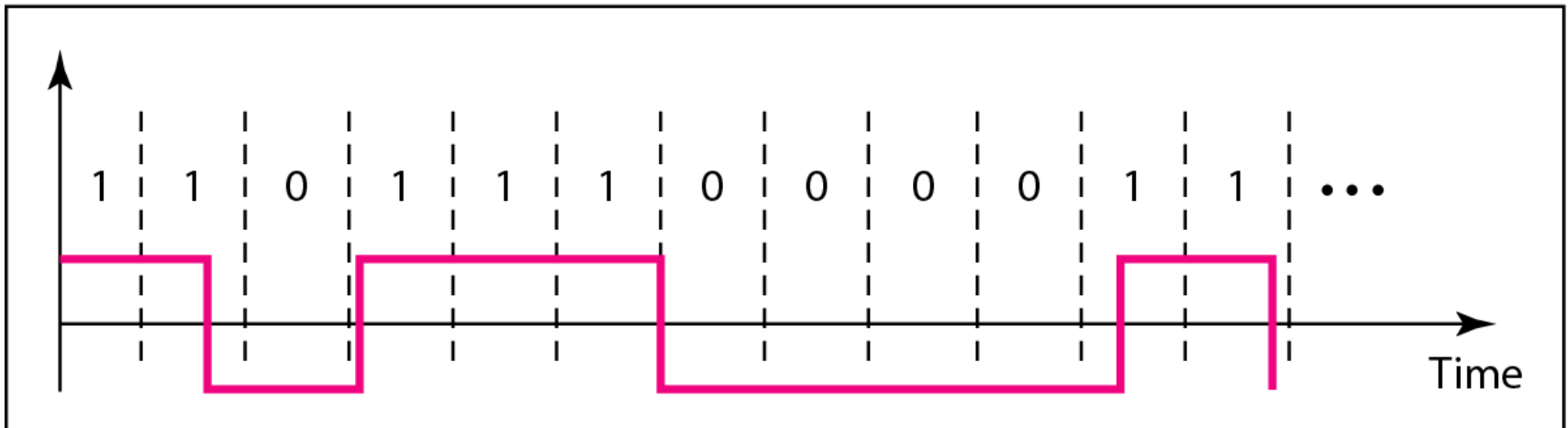


b. A signal without dc component

Lack of Synchronization

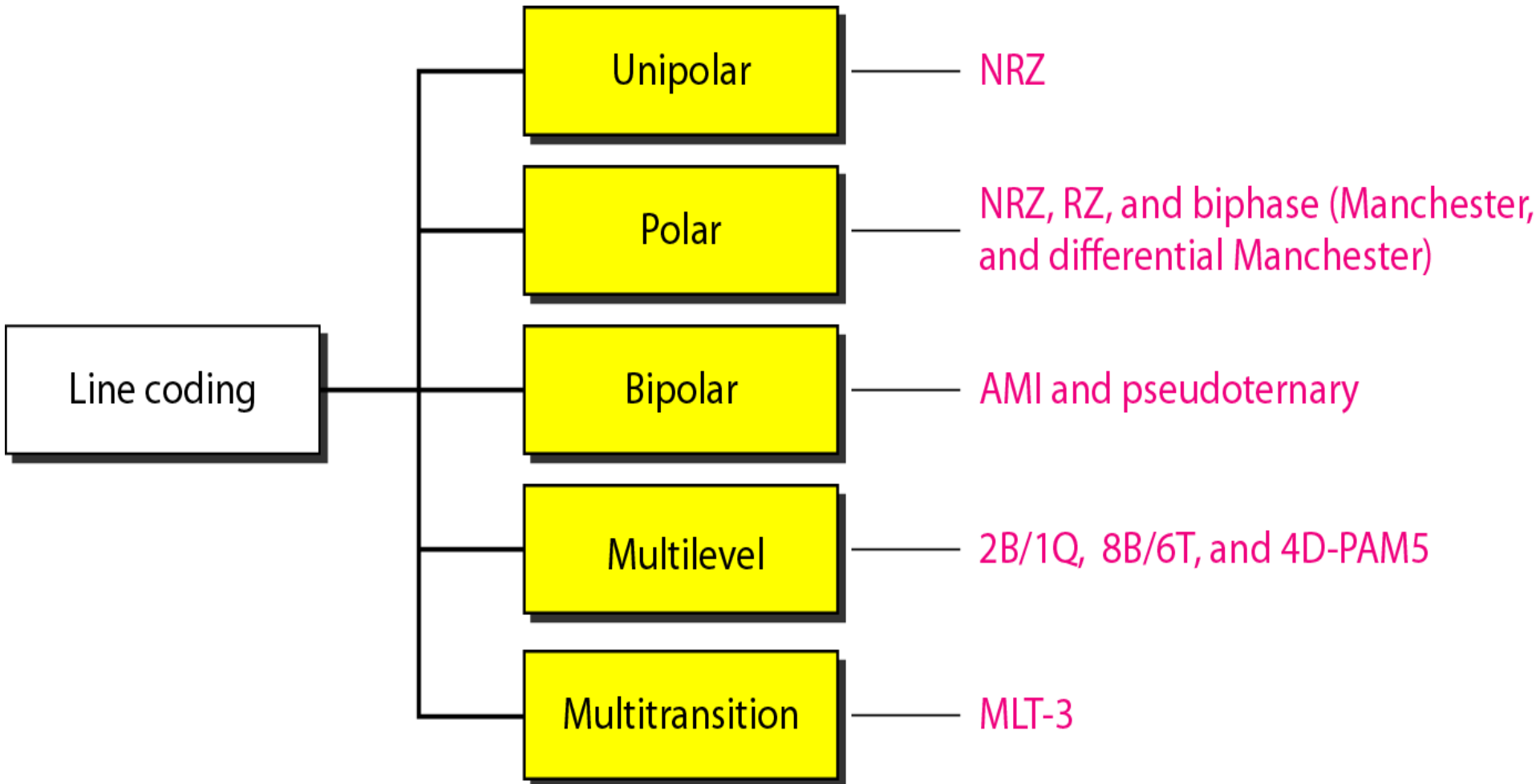


a. Sent



b. Received

Line Coding

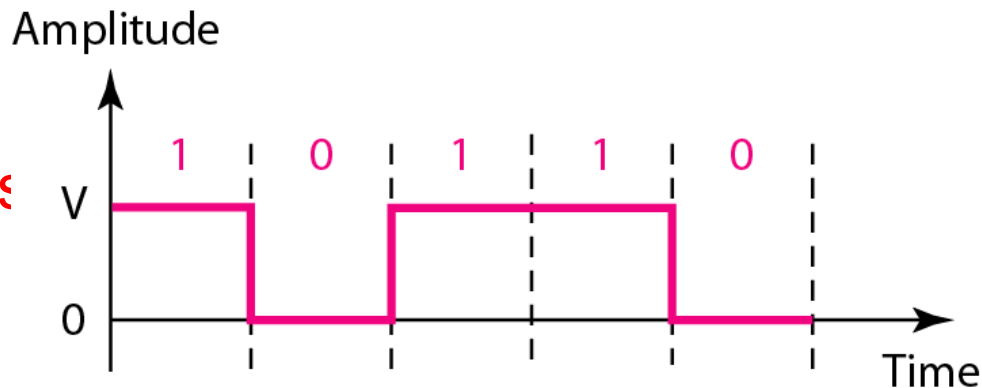


Unipolar

In **Unipolar scheme**, all the signal levels are **one side** of the time axis.

- It is designed as a **Unipolar Non-Return-to-Zero (Unipolar NRZ)** scheme
- A +ve voltage represents a 1 and zero voltage represents a 0. *The signal does not return to zero at the middle of a bit*

- Requires



$$\frac{1}{2}V^2 + \frac{1}{2}(0)^2 = \frac{1}{2}V^2$$

Normalized power

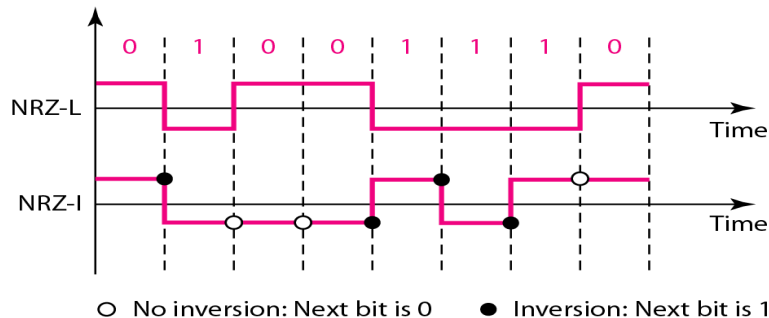
Polar

In **Polar schemes**, the signal levels are **both** sides of the time axis.

- Designed as a **Polar NRZ, RZ, and Biphase (Manchester and Differential Manchester)** schemes
- In **Polar Non-Return-to-Zero (Polar NRZ)**, **two** voltage levels (one + and one -) are used, for a 1 and a 0
 - Polar NRZ can be of two types: **NRZ-Level (NRZ-L)** or **NRZ-Invert (NRZ-I)**

Polar NRZ

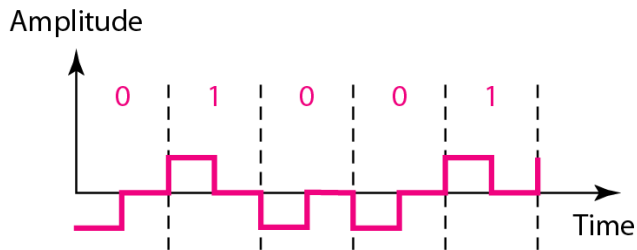
- In Polar **NRZ-Level (NRZ-L)**, the level of the voltage determines the bit
- In Polar **NRZ-Invert (NRZ-I)**, the change or lack of change of voltage level determines the bit
 - For a 0 there is no change in the level; for a 1 there is



- Both schemes have an average signal rate of $N/2$ baud
- Both have DC component problem

Polar Return-to-Zero (RZ)

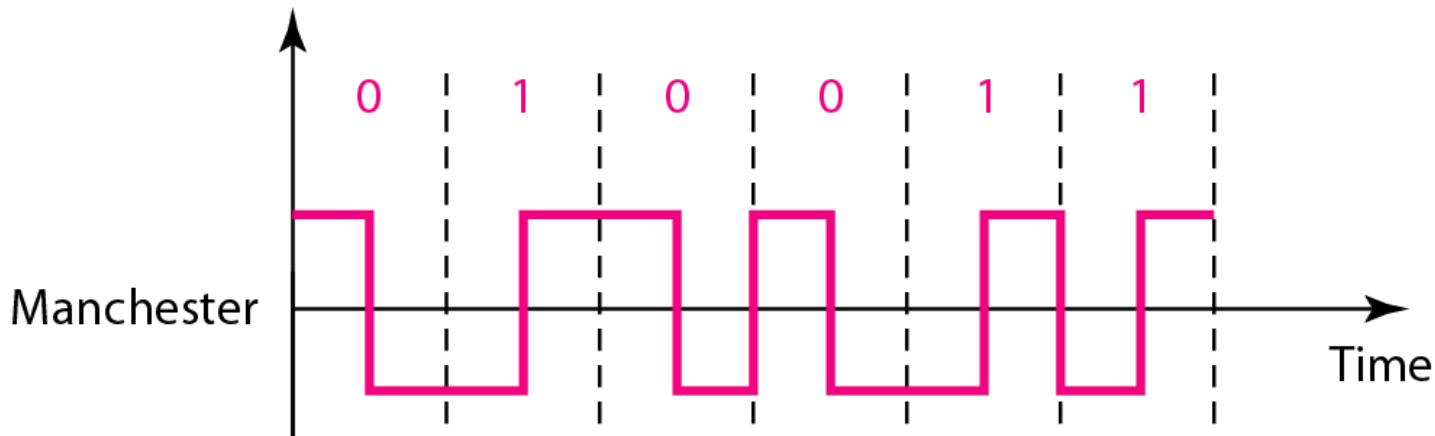
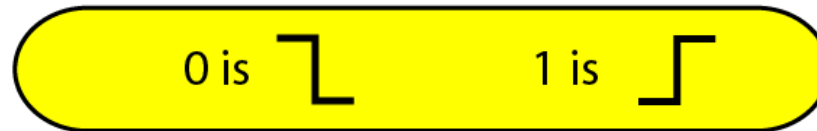
- **Uses 3 levels: Negative, Zero, and Positive**
- **Signal level goes to 0 at the middle of every bit; At the beginning of next bit, the level changes**



- Requires two signal changes for every bit. Hence more bandwidth is required
- Creating three voltage levels is more complex
- Hence the scheme is not preferred

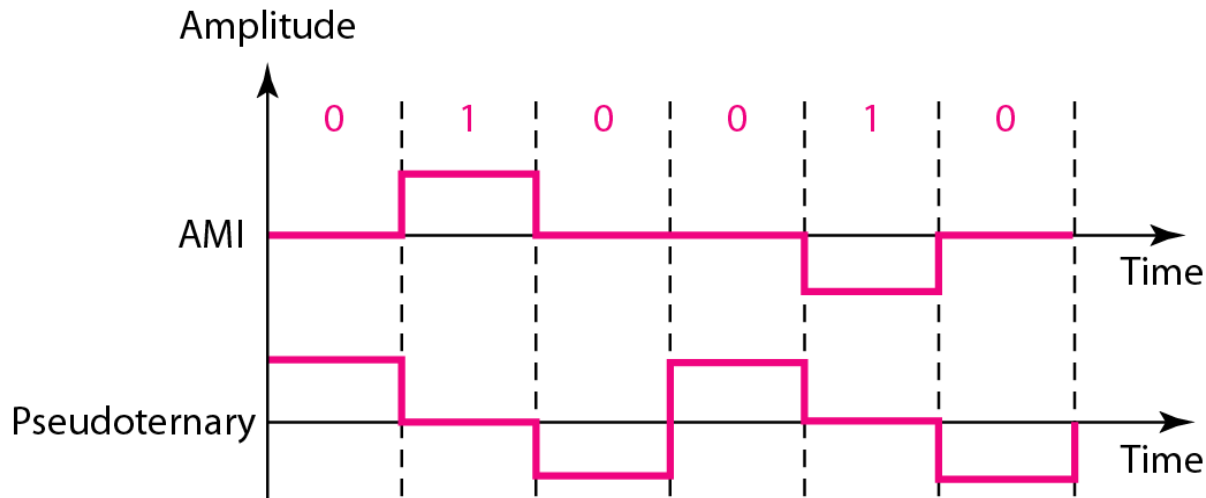
Biphase – Manchester Encoding

- Bit duration is divided into two halves. For a 0, the signal is +ve during the first half and changes to -ve in the second half. For a 1, it is reverse
- Combines the ideas of RZ and NRZ-L



Bipolar Schemes...

- Use **3 levels**: **Negative**, **Zero**, and **Positive** (like Polar RZ)
- **2 variations**: **Alternate Mark Inversion (AMI)** and **Pseudoternary**
 - 1. AMI**: 0's are represented as 0 voltage and 1's are represented by alternating +ve and -ve voltages
 - 2. Pseudoternary**: 0's are represented by alternating +ve and -ve voltages and 1's are represented by 0 voltage



Bipolar Schemes

- Has the same signal rate as NRZ
- But there is no DC component, because of **alternations for the same bit**
- **AMI** is used for long distance communication
 - Has **synchronization problem** when there is a long sequence of 0's, but this is overcome by **scrambling**

Multilevel Schemes

A pattern of **m data elements** is encoded as a pattern of **n signal elements**

- **2 bit values (0 or 1) and m data bits in each group - can have 2^m combinations**
- If we have **L** signal values (levels) and **n** signal elements in each group, there can be **L^n combinations (patterns)**
 - If **$2^m = L^n$** , for each data **combination** - one signal **pattern**
 - If **$2^m < L^n$** , only a **subset of all possible patterns** is used
 - If **$2^m > L^n$** , encoding is **not possible**

Multilevel Schemes...

- **Multilevel coding** is termed **mBnL**
 - m - Length of the binary pattern (No. of bits)
 - B - Binary
 - n - Length of the signal pattern
(Number of pulses in each group of pulses)
 - L - Number of levels - A letter is used in place of L
 - 2 – B (Binary)
 - 3 – T (Ternary)
 - 4 – Q – Quaternary

E.g. **2B1Q** – 2 Binary, 1 Quaternary

$$m=2, n=1, L=4$$

Data patterns of size 2 are encoded as one signal element

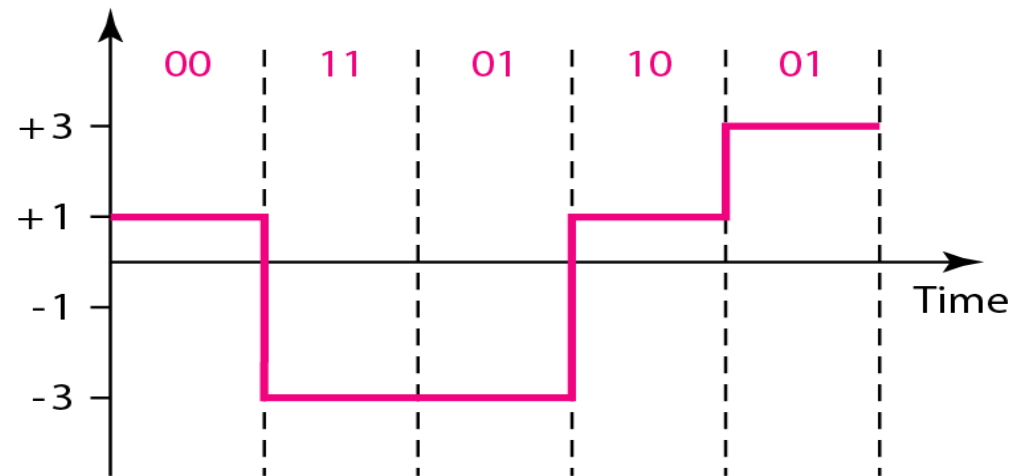
Note: Here $2^m = L^n$

2B1Q - Multilevel Scheme

Previous level: positive Previous level: negative

Next bits	Next level	Next level
00	+1	-1
01	+3	-3
10	-1	+1
11	-3	+3

Transition table



Assuming positive original level

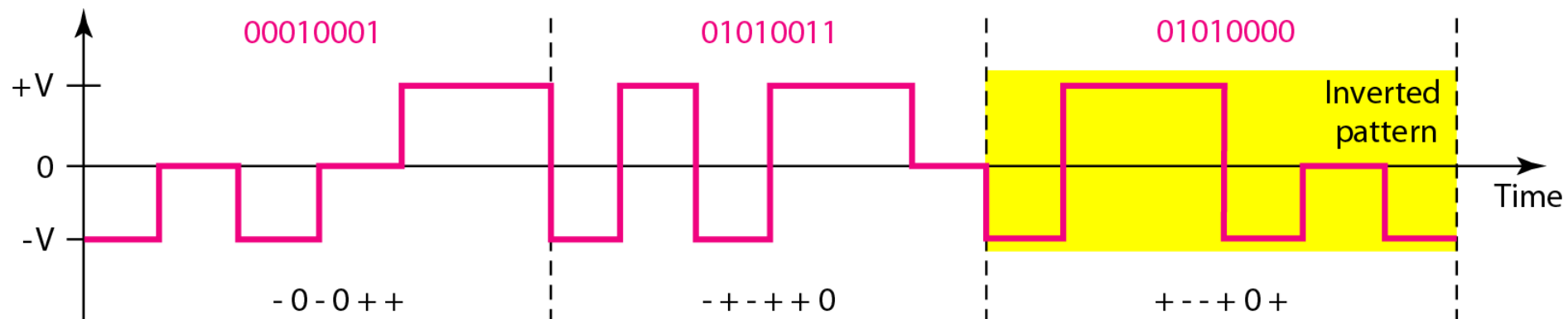
Used in Digital Subscriber Line (DSL) - provides high speed internet connection using subscriber telephone lines

8B6T Scheme

8B6T– 8 Binary, 6 Ternary

- $m=8, n=6, L=3$, Hence $2^m < L^n$
- 8-bit patterns are encoded as a 6-signal elements pattern
- Each signal pattern has a weight of 0 or +1; There is no pattern with weight -1
- If two patterns of weight +1 are encountered in succession, the second is totally inverted to give a weight of -1

Refer Mapping Table in
Appendix D - Text

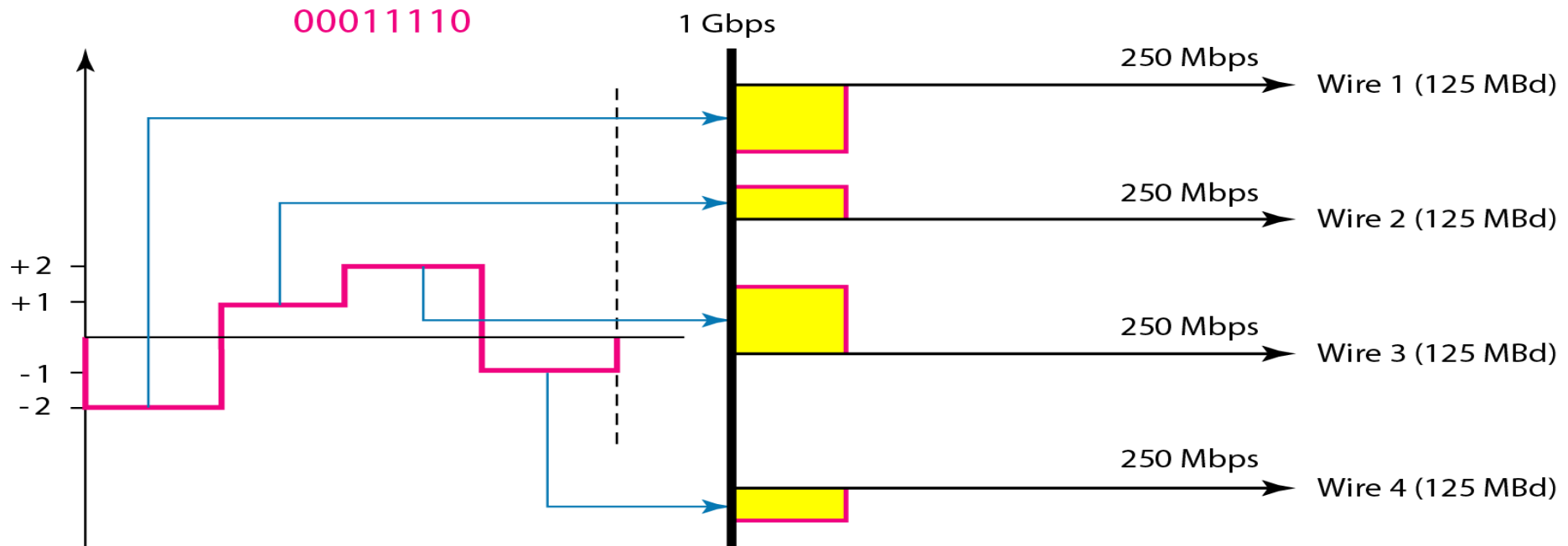


Used with 100BASE-4T cable

4D-PAM5 Scheme

4D-PAM5: 4-Dimensional, 5-level Pulse Amplitude Modulation

- Data is sent over 4 wires at the same time
- Uses 5 voltage levels (-2,-1,0,1,2): **0 level for Error detection only**



Used in Gigabit LANs

- Data is sent on 4 wires that can handle 125 Mbaud

Multi line Transmission: MLT-3

- **Multi Line Transmission**

- Uses 3 levels – (+v, 0, -v)

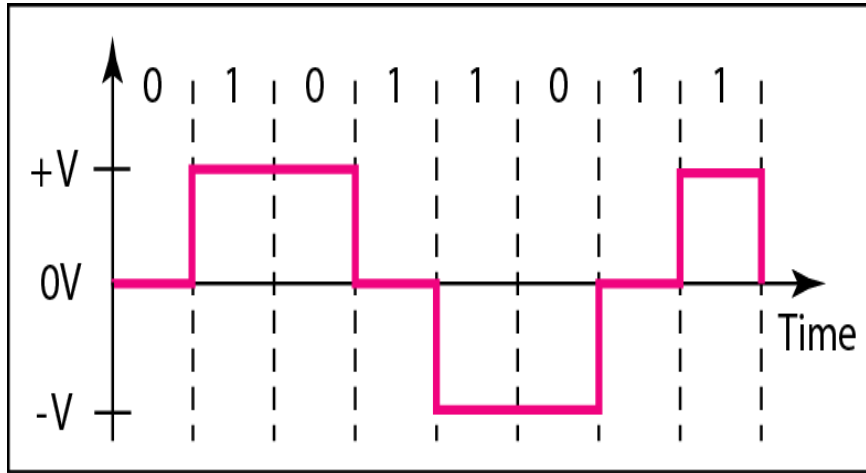
- **3 transition rules**

1. If the next bit is **0**, there is no transition
2. If the next bit is **1** and the current level is not **0**, the next level is **0**
3. If the next bit is **1** and the current level is **0**, the next level is the opposite of the nonzero level

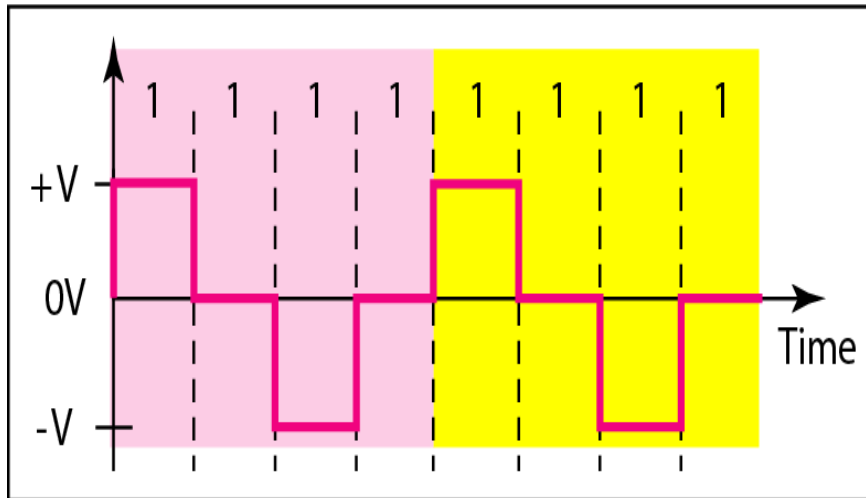
- Signal rate is $\frac{1}{4}$ of the bit rate

- Hence we can achieve a transmission rate of 100 Mbps on a copper wire that cannot support more than 32 MHz

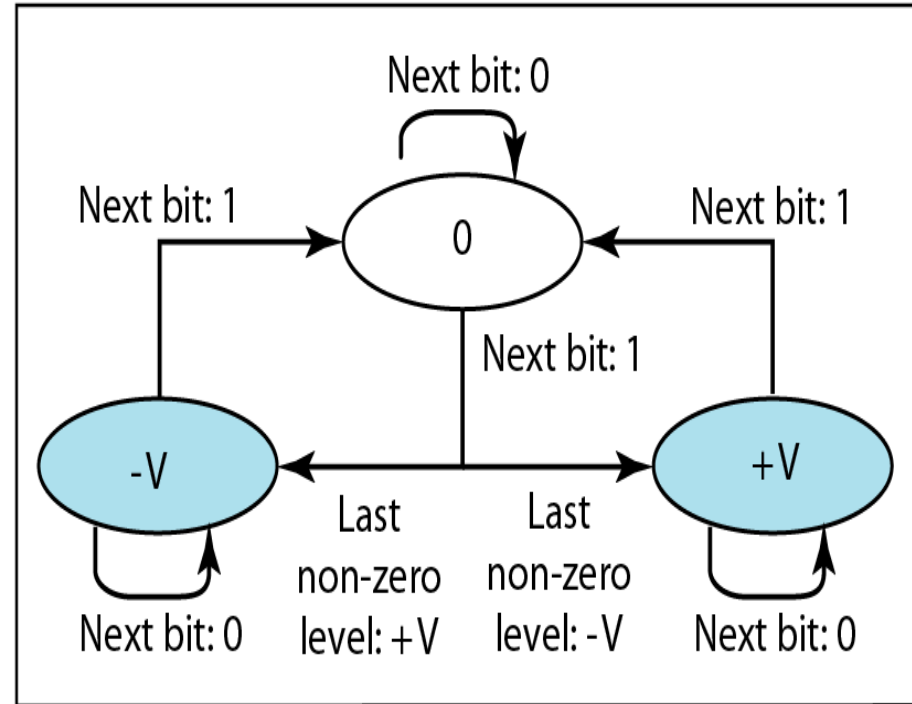
MultiLine Transmission: MLT-3



a. Typical case



b. Worse case



c. Transition states

1. During online examinations, answer scripts are transmitted continuously. The network administrator wants a coding scheme that reduces the DC component and provides error detection capability. Which line coding scheme would you recommend?

Answer: AMI (Alternate Mark Inversion) because it eliminates the DC component and helps detect transmission errors through bipolar violations.

2. A university is upgrading its campus network to support smart classrooms, cloud-based laboratories, HD video lectures, and IoT-enabled devices. The network administrator wants a line coding technique that can transmit large amounts of data efficiently while minimizing bandwidth consumption. Which category of line coding would you recommend?

Answer: Multilevel Coding

3. A multi-specialty hospital has deployed hundreds of IoT-enabled patient monitoring devices in its Intensive Care Unit (ICU). These devices continuously transmit critical patient information such as heart rate, blood pressure, oxygen saturation (SpO_2), ECG signals, and respiratory rate to a centralized monitoring station. Since patient safety depends on real-time and accurate data transmission, the hospital network must maintain reliable synchronization even during long periods of continuous data transfer. As a network engineer, recommend the most suitable line coding scheme and justify your choice based on synchronization requirements, bandwidth efficiency, and communication reliability.

Answer: Manchester Encoding (Bi phase Coding)

4. A national bank connects all its branches through a high-speed network. The bank transfers large volumes of customer records and transaction data daily. Since bandwidth is limited and efficient utilization is important, which category of line coding would you recommend?

Answer:

Multilevel Coding because it represents multiple bits using a single signal element, improving bandwidth utilization and supporting high-speed communication.